

What Is Time? A Deterministic Interpretation Under the MMA-DMF Framework

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Abstract

Time is modeled in MMA-DMF as local causal relaxation of an active, memory-bearing vacuum. Clock rates are local screened or unscreened vacuum-response rates. Einstein and general-relativistic timing and geometric observables are recovered as dense screened Einstein-limit projections rather than discarded. The temporal formulation is continuous with prior MMA-DMF relativity, gravitoelectromagnetism, and high-energy manuscripts [1, 2, 3]. The present manuscript synthesizes that temporal sector by stating the ontology of time, giving the equation spine, organizing completed convergence and bridge layers, and separating the remaining decisive external falsification gates from completed formulation. Completed fixed-rule analyses and public-data convergence/bridge layers include Bell/NIST CH-Eberhard closure, a derived CHSH bridge, high-energy FRB/magnetar/NASA Archives/NuSTAR/IXPE timing sector, GNOME public source products, Voyager LEC/PDS boundary context, Technion/topological phase dynamics, GNSS source products, HETDEX/Ly α bridge, and mathematical consistency of the stated equations. The remaining open items are not missing theory support. They are decisive external falsification gates: NASA/Voyager DSN operational or radiometric residual records and GNOME Science Run 2 station-level timing streams. The missing NASA/Voyager and GNOME records are not required to formulate the MMA-DMF temporal ontology; they are required only for independent external falsification of the strongest boundary-timing predictions.

Keywords: MMA-DMF; time ontology; active vacuum; memory kernel; fluctuation–dissipation theorem; local clocks; Einstein-limit recovery; Bell contextuality; fast radio bursts; GNOME; Voyager; GNSS; heliopause; negative chirp.

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1 Introduction

Quantum mechanics normally uses time as the parameter against which state evolution is written, whereas general relativity makes clock rates local, path-dependent, and sensitive to the metric environment [4, 5, 6]. The resulting problem is not only technical. It is ontological: is time a universal parameter, a geometric coordinate, a thermodynamic ordering, an operational clock reading, or something deeper?

This manuscript synthesizes the temporal sector of the MMA-DMF framework as developed across the prior MMA-DMF relativity, gravitoelectromagnetism, and high-energy formulations [1, 2, 3]. Its purpose is to state the ontology of time, connect it to the existing MMA-DMF equation family, organize completed convergence layers, and identify the remaining decisive falsification gates. The present paper does not depend on unreleased NASA or GNOME records for formulation. Those records are external adjudication layers.

The core proposal is direct. Within MMA-DMF, time is not treated as a rigid background metronome and not treated as an empty geometric container. Time is modeled as the local causal relaxation of an active vacuum that carries memory, dissipates perturbations, and responds differently in dense screened environments and low-density boundary environments. A clock is then a local response system. It ticks according to the screened or unscreened relaxation state of the physical vacuum surrounding and dressing its constituents. This is why the paper can affirm the Einstein limit while still asking whether that limit is the final ontology of time.

The phrase *Einstein limit* is used carefully. It does not mean that standard relativistic clock phenomenology is ignored. It means that, in dense screened conditions, MMA-DMF must recover the same operational timing behavior that makes relativity indispensable in GPS/GNSS, Shapiro timing, orbital mechanics, gravitational redshift and local Lorentz kinematics [7]. The framework's claim is not that known relativistic clocks are wrong. Its claim is that their successful geometric description can be interpreted as the screened projection of deeper local vacuum-response dynamics.

The paper is organized around four levels. First, the formal equation spine defines the theory's temporal sector. Second, the ontology interprets the equations: retarded memory supplies temporal direction, density-dependent relaxation supplies local time, and fluctuation–dissipation closure supplies thermodynamic stability. Third, completed convergence and bridge layers are placed in a hierarchy rather than listed as one undifferentiated dossier. Fourth, the strongest boundary-timing predictions are left where they belong: at the external falsification gate.

This structure is intentionally different from a broad technical inventory. The manuscript keeps the central thesis visible. FRB, Bell, GNOME, Voyager, GNSS, HETDEX, Technion and gravitational-wave echo screens are not all made to carry the same evidential weight. Bell and high-energy timing are completed convergence or bridge layers under explicit boundaries. GNOME public products and Voyager LECP/PDS products are strong public and boundary context layers. The decisive tests still require station-level GNOME timing streams and NASA/Voyager DSN records. That distinction strengthens the manuscript because it prevents both overclaiming and underclaiming.

2 Claim Boundary and Evidence Classification

The title asks a strong question, but the answer must be bounded. The paper therefore uses the evidence classification in Table 1. The central formal claim is the definition of time inside MMA-DMF. The convergence layers establish continuity, closure or bridge behavior. The remaining NASA/Voyager and GNOME records are not needed to state the framework; they are needed to decide whether the strongest boundary-timing predictions survive independent external adjudication.

Table 1: Evidence classes used in the manuscript. The classification separates model definition, interpretation, completed convergence and decisive external falsification.

Class		Meaning	Allowed use	Typical location
Core formal claim		A statement that follows from the stated MMA-DMF equations or definitions.	May define the framework and appear in the abstract if written as “within MMA-DMF”.	Abstract; mathematical core
Model-supported interpretation	in-	A physical interpretation supported by the equation spine but not by itself an external proof.	May be used in the body with equation links and falsification routes.	Ontology; discussion
Internal mathematical consistency		A consistency result for the stated equations, limits, derivatives, kernels or status logic.	May support closure of the formal spine; not a substitute for external data.	Mathematical core; appendix
Completed convergence or bridge layer	conver-	A public-data, theoretical or computed layer retained with a closed status and explicit boundary.	May be presented as a completed convergence layer, not as final proof of the ontology.	Results; convergence section
Public-data layer	source	A layer based on public products that supports detector landscape, sensitivity, ephemeris or source context.	May constrain interpretation and identify gates, but cannot replace missing station or residual streams.	GNOME, Voyager, GNSS
Decisive external falsification gate		A required record set whose release or analysis can independently support or weaken the strongest boundary-timing predictions.	May be listed as pending without weakening the formulation of the theory.	Remaining gates
Speculative extension/future work	exten-	An interpretive or exploratory branch not required for the temporal ontology.	Must not control the abstract or central conclusion.	Discussion; appendix

Table 2: Claim-boundary table used throughout the manuscript.

Claim	Allowed wording	Not allowed	Evidence class	Location in paper
Time ontology	Within MMA-DMF, time is modeled as local causal relaxation of an active, memory-bearing vacuum.	MMA-DMF proves the final nature of time.	Core formal claim	Abstract; Sections 1–5
Arrow of time	The retarded kernel and FDT closure make temporal directionality part of the model.	The second law is replaced or experimentally proved.	Core formal claim; model-supported interpretation	Sections 5–6
No master clock	The framework predicts local density-dependent timing response; station-level GNOME data are a decisive gate.	Existing public GNOME products already prove density-delay timing.	Model-supported interpretation; pending external falsification	Sections 7, 10–11
Clock recovery	Dense screened regimes recover Einstein-limit clock observables as an effective projection.	GR clock tests are invalid or unnecessary.	Model-supported interpretation; completed convergence layer	Section 8
Bell sector	CH/Eberhard is retained as a strong external Bell layer; the all-trials CHSH bridge is derived and bounded.	The derived bridge is a NIST-published 5σ CHSH result.	Completed bridge layer; computed bridge layer	Section 10

Claim	Allowed wording	Not allowed	Evidence class	Location in paper
FRB/high-energy sector	FRB sign asymmetry, magnetars, NASA Archives, NuSTAR and IXPE products form a high-energy convergence sector.	FRBs uniquely prove MMA-DMF or exclude conventional astrophysics.	Completed convergence layer	Section 10
Voyager LECP/PDS	LECP and PDS/SPICE/SEDR products give strong boundary and ephemeris context.	LECP/PDS are DSN observed-minus-computed residual proof.	Completed context layer; decisive gate pending	Section 10–11
GNOME public products	Public source products support sensitivity, noise and ALP/domain-wall bound layers.	Public products are station-level negative-chirp validation.	Public-data convergence layer; decisive gate pending	Section 10–11
Cyclic or subjective extensions	These are interpretive or future-work extensions.	They are necessary to formulate the temporal ontology.	Speculative extension/future work	Discussion and appendix

2.1 Narrative hierarchy

The strongest version of the paper does not treat every test as an equal proof attempt. The hierarchy used here is:

1. **Temporal ontology:** time is modeled as local causal relaxation of an active, memory-bearing vacuum.
2. **Equation spine:** generalized Langevin dynamics, density-dependent effective mass, retarded and even kernels, FDT closure, geometric relaxation and density-delay relations.
3. **Einstein-limit recovery:** ordinary clock and geometric observables are recovered as dense screened projections.
4. **Completed convergence and bridge layers:** Bell, high-energy timing, GNOME public products, Voyager LECP/PDS context, Technion/topological phase dynamics, HETDEX/Ly α , GNSS source products and GW echo screens.
5. **Remaining decisive falsification gates:** NASA/Voyager DSN operational or radiometric residual records and GNOME Science Run 2 station-level timing streams.

This hierarchy prevents a reader from mistaking a broad convergence map for a single validation claim.

3 MMA-DMF Temporal Ontology

3.1 The answer in one sentence

Within MMA-DMF, time is the local retarded update of the active vacuum. This statement has three parts. *Local* means that timing is generated by the state of the environment, not by a universal master clock. *Retarded* means that the physical response depends on the past light-cone and the causal history of the vacuum, not on advanced response. *Update* means that time is the physically realized ordering of state changes, not an independent substance.

The active-vacuum and gravitoelectromagnetic interpretation follows the prior MMA-DMF gravitoelectromagnetism formulation and the associated relativity closure [2, 1]. The framework

therefore occupies a position between two familiar extremes. It does not preserve Newtonian absolute time as fundamental. It also does not make the metric the final ontological layer. In dense screened conditions, the metric description is recovered as an effective projection. In boundary or weakly screened regimes, the framework predicts that residual timing, relaxation or contextual signatures can expose the deeper response layer.

3.2 Clock rates as response rates

A clock is a physical oscillator. Atomic transition clocks, pulsar timing, spacecraft radiometric tracking and laboratory interferometers all depend on material and field degrees of freedom. In MMA-DMF, those degrees of freedom are dressed by the local scalar-vacuum response. A clock rate is therefore not an abstract flow of time. It is the local rate at which the clock's physical constituents relax, oscillate and phase-accumulate under the environment-dependent response of the vacuum.

Dense environments produce screened behavior. In those regimes the response is stiff and the Einstein-limit projection is recovered. Low-density or transition regions can become more sensitive to boundary terms, delayed relaxation or achromatic residuals. The ontology therefore does not deny relativistic clocks. It gives them a mechanism: clocks tick as local vacuum-response systems.

3.3 Arrow of time and memory

The arrow of time is treated as more than a coarse-grained entropy slogan. The theory uses a retarded kernel in the dynamical equation. Because the kernel enters as a causal memory integral, the future state depends on past deformations, while the reverse dependence is not present. The even kernel used for equilibrium FDT normalization is not a second causal channel. It is the autocorrelation object required to normalize noise consistently.

This distinction is central. Without it, the model would either risk acausal interpretation or lose thermodynamic closure. With it, the arrow of time is tied to the same equations that define stochastic response and dissipative relaxation.

4 Mathematical Core

The equation set used here follows the MMA-DMF formulation established in previous MMA-DMF relativity and high-energy manuscripts [1, 2, 3]. Equations are referred to only by manuscript equation numbers.

4.1 Symbols, units, and conventions

Unless otherwise stated, the equations use natural units with $c = \hbar = 1$ and retain k_B explicitly in thermodynamic expressions. Length and time therefore carry inverse mass dimension. The symbols most likely to raise dimensional questions are collected in Table 3. In SI-language interpretations, τ is converted back to seconds and M to an energy scale in the usual way.

Table 3: Symbols, units, and dimensional conventions used in the temporal equation spine. Natural units $c = \hbar = 1$ are used unless stated otherwise; k_B is kept explicit in FDT expressions.

Symbol	Natural-unit dimension	di- Operational meaning	Notes / equation links
M	mass / energy	Fundamental vacuum rigidity scale used in response, kernel, and cutoff terms.	Appears in m_{eff} , kernels, and relaxation rates; see Eq. (2), Eq. (5), and Eq. (9).
ρ	mass ⁴	Local or effective environmental energy density.	Point-density inputs are replaced by Yukawa-smoothed ρ_{eff} in Eq. (3)–(4).
ρ_{sat}	mass ⁴	Saturation density controlling the transition from unsaturated to screened relaxation.	Enters the full relaxation rate in Eq. (9); the low-density delay law Eq. (12) assumes $\rho \ll \rho_{\text{sat}}$.
$A(\rho_{\text{env}}, G)$	mass ²	Time-domain memory-kernel amplitude.	In Eq. (5) and Eq. (6), A sets the strength of retarded dissipation and even autocorrelation.
T_{vac}	temperature; $k_B T_{\text{vac}}$ has mass dimension	Effective vacuum-bath temperature used in fluctuation–dissipation normalization.	Appears in Eq. (7) and Eq. (8); it is not an independent timing parameter in this manuscript.
η_{drag}	dimensionless	Drag normalization entering the regulated stochastic-force spectrum.	Used in Eq. (8); density dependence is carried through the environmental response.
$\Gamma_{\text{ret}}(\Delta t)$	mass ²	Causal time-domain memory kernel.	Appears inside the GLE memory integral in Eq. (1); its Fourier transform has one lower mass dimension.
$\Gamma_{\text{geo}}(\rho)$	mass	Local geometric relaxation rate.	Defines local response time through $\tau = \Gamma_{\text{geo}}^{-1}$ in Eq. (9).
$\tau(\rho)$	mass ^{−1} / time	Local relaxation time, converted to seconds for observational timing.	The full differential delay is Eq. (11); Eq. (12) is the low-density approximation.

4.2 Generalized Langevin equation

The temporal dynamics are organized by the relativistic generalized Langevin equation

$$\frac{1}{c_s^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi + [m_{\text{eff}}^{\text{ren}}(\rho_{\text{eff}})]^2 \phi + \int_{-\infty}^t \Gamma_{\text{ret}}(t - t'; \rho_{\text{env}}, G) \dot{\phi}(t') dt' = F_{\text{noise}}(\mathbf{x}, t). \quad (1)$$

The first term enforces hyperbolic propagation; the mass term encodes environmental response; the memory integral makes the dynamics non-Markovian; and the stochastic force closes the quantum-statistical bath. Eq. (1) is the temporal engine. It does not by itself prove the ontology, but it defines the dynamical object whose interpretation is time as local physical update.

4.3 Density-dependent mass and response length

The density-dependent mass and associated response length are

$$m_{\text{eff}}(\rho) = \frac{\sqrt{\rho}}{M}, \quad m_{\text{eff}}^2(\rho) = \frac{\rho}{M^2}, \quad \lambda_\phi(\rho) = \frac{1}{m_{\text{eff}}(\rho)} = \frac{M}{\sqrt{\rho}}. \quad (2)$$

Eq. (2) is the compact expression behind screening, density-dependent relaxation, clock modulation and local time. Large density yields large effective mass and short response length. Low density yields longer response and greater sensitivity to boundary timing.

The scalar does not respond to a singular point-density input. The effective density used in timing and screening contexts is the normalized Yukawa-smoothed density

$$\rho_{\text{eff}}(\mathbf{x}) = \int_{\mathbb{R}^3} \rho_{\text{matter}}(\mathbf{y}) K_Y(|\mathbf{x} - \mathbf{y}|; \lambda_{\text{cut}}) d^3y, \quad (3)$$

where

$$K_Y(r; \lambda_{\text{cut}}) = \frac{1}{4\pi\lambda_{\text{cut}}^2 r} \exp\left(-\frac{r}{\lambda_{\text{cut}}}\right), \quad \int_{\mathbb{R}^3} K_Y(r; \lambda_{\text{cut}}) d^3r = 1. \quad (4)$$

The normalization in Eq. (4) is important because station-level density response must be a physical convolution, not an arbitrary point insertion.

4.4 Retarded memory and even FDT kernel

The causal kernel used in the memory integral is

$$\Gamma_{\text{ret}}(\Delta t) = A(\rho_{\text{env}}, G) e^{-M\Delta t} \Theta(\Delta t), \quad \tilde{\Gamma}_{\text{ret}}(\omega) = \frac{A}{M + i\omega}, \quad \text{Re } \tilde{\Gamma}_{\text{ret}}(\omega) = \frac{AM}{M^2 + \omega^2}. \quad (5)$$

The equilibrium autocorrelation kernel used in FDT normalization is

$$\Gamma_{\text{even}}(\Delta t) = A(\rho_{\text{env}}, G) e^{-M|\Delta t|}, \quad \tilde{\Gamma}_{\text{even}}(\omega) = \frac{2AM}{M^2 + \omega^2}. \quad (6)$$

The factor of two in Eq. (6) belongs to the even autocorrelation spectrum. It is not interpreted as an advanced causal response. This separation is one of the mathematical safeguards that keeps the arrow-of-time interpretation clean.

4.5 Fluctuation–dissipation closure

For equilibrium vacuum temperature T_{vac} , the noise covariance is closed by

$$\langle F_{\text{noise}}(\omega) F_{\text{noise}}(\omega') \rangle = 4\pi k_B T_{\text{vac}} \text{Re } \tilde{\Gamma}_{\text{ret}}(\omega) \delta(\omega + \omega'). \quad (7)$$

Equivalently, a regulated spectral form can be written as

$$S_F(k, \omega) = 4k_B T_{\text{vac}} \eta_{\text{drag}}(\rho_{\text{env}}, G) \frac{M^2}{M^2 + \omega^2} \exp\left(-\frac{k^2}{M^2}\right). \quad (8)$$

Equations (7) and (8) are the thermodynamic closure. They prevent the stochastic sector from becoming an arbitrary source of energy injection and connect vacuum noise to dissipation [8, 9, 10, 11].

4.6 Geometric relaxation rate and local time

The density-dependent geometric relaxation rate is

$$\Gamma_{\text{geo}}(\rho) = \frac{\sqrt{\rho}}{M} \left[1 + \left(\frac{\rho}{\rho_{\text{sat}}} \right)^2 \right]^{-1/4}, \quad \tau(\rho) = \Gamma_{\text{geo}}^{-1}(\rho). \quad (9)$$

A useful derivative identity is

$$\frac{d\Gamma_{\text{geo}}}{d\rho} = \frac{1}{2M\sqrt{\rho}} \left[1 + \left(\frac{\rho}{\rho_{\text{sat}}} \right)^2 \right]^{-5/4}, \quad \frac{d\tau}{d\rho} = -\Gamma_{\text{geo}}^{-2} \frac{d\Gamma_{\text{geo}}}{d\rho} < 0. \quad (10)$$

The sign in Eq. (10) is the local-time statement in differential form: as density increases, the relaxation time decreases.

The full station-to-station density-delay relation is

$$\Delta t_{ij}^{\text{full}} = \tau(\rho_i) - \tau(\rho_j), \quad \tau(\rho) = \Gamma_{\text{geo}}^{-1}(\rho), \quad (11)$$

with $\Gamma_{\text{geo}}(\rho)$ defined in Eq. (9). In the unsaturated or low-density regime $\rho_i, \rho_j \ll \rho_{\text{sat}}$, Eq. (11) reduces to

$$\Delta t_{ij} = \tau_i - \tau_j = \tau_{\text{ref}} \left[\left(\frac{\rho_i}{\rho_{\text{ref}}} \right)^{-1/2} - \left(\frac{\rho_j}{\rho_{\text{ref}}} \right)^{-1/2} \right]. \quad (12)$$

Eq. (12) is therefore an unsaturated approximation of Eq. (11), not a replacement for the saturated relation in Eq. (9). It is the mathematical form of the no-master-clock thesis for a sensor network in the low-density approximation. It does not say that every public GNOME product already validates density-delay timing. It says what station-level streams would need to test.

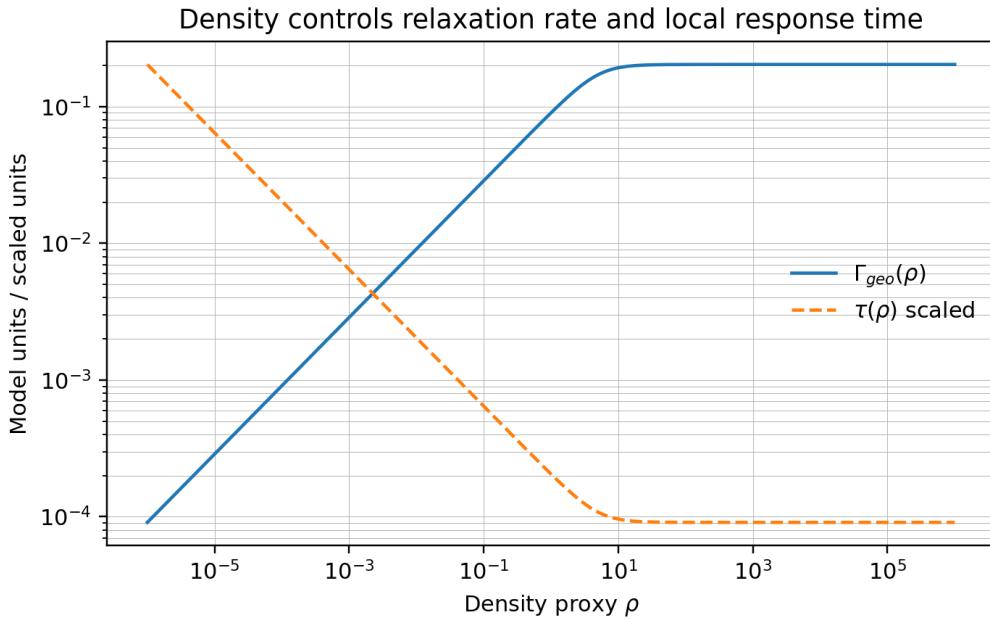


Figure 1: Density-to-relaxation mapping in the MMA-DMF temporal sector. The plotted structure illustrates how $\Gamma_{\text{geo}}(\rho)$ and $\tau(\rho)$ turn environmental density into local response time, supporting Eq. (9), Eq. (11) and Eq. (12).

4.7 Negative chirp and network coherence

A causal dissipative relaxation template can be written as

$$\nu(t) = \nu_0 \left[1 - k \left(1 - e^{-t/\tau} \right) \right], \quad \frac{d\nu}{dt} = -\frac{\nu_0 k}{\tau} e^{-t/\tau} < 0. \quad (13)$$

The frequency derivative in Eq. (13) is the operational sign of dissipative relaxation. A positive chirp is not automatically impossible in all astrophysics, but within this MMA-DMF template a robust, globally coherent scalar-like positive chirp surviving controls would be a direct stress test.

For a network, the coherence functional can be written schematically as

$$\mathcal{L}_{\text{net}} = \prod_i p[d_i(t) \mid A_i h(t - t_0 - \Delta t_i(\rho_i); \tau_i) + n_i(t)], \quad (14)$$

where $d_i(t)$ is the station record, h is the negative-chirp template, Δt_i follows Eq. (12), and n_i denotes local noise. The GNOME station-level gate is the test of whether Eq. (14) beats flat simultaneity under blind controls.

Table 4: Core MMA-DMF temporal equation spine. Each row links the role of a formula family to the manuscript equation number, prior MMA-DMF continuity, status, and falsification route.

Equation family	Manuscript formula link	Role in time ontology	Prior MMA-DMF ref-erence	Status	Falsification route
Relativistic GLE	(1)	Defines causal local update of the active vacuum.	[2, 1, 3]	Core claim	formal Failure of causal, stable dynamics under the stated kernel and FDT closure.
Density-dependent mass	(2)	Converts local density into local response length and local clock-rate modulation.	[1, 3]	Core claim	formal Precision density-modulation tests showing no response where the model predicts one.
Yukawa smoothing	(4)	Defines the effective density seen by the scalar sector rather than a point-density singularity.	[2, 1]	Core claim	formal Failure of normalized effective-density construction or incompatible station-level density response.
Retarded kernel	(5)	Places the arrow of time in causal memory rather than in a symmetric response.	[2, 1]	Core claim	formal Observation of admissible positive-chirp scalar transients under the same selection rules.
Even/FDT kernel	(6)	Supplies the equilibrium autocorrelation object without adding advanced response.	[2]	Core claim	formal Thermodynamic inconsistency or factor-of-two ambiguity in reproducible calculations.

Equation family	Manuscript formula link	Role in time ontology	Prior MMA-DMF reference	ref-Status	Falsification route
FDT closure	(7)	Locks noise power to dissipation and prevents arbitrary stochastic forcing.	[2, 3]	Core formal claim	Runaway scalar heating, non-normalizable noise, or incompatible equilibrium response.
Geometric relaxation	(9)	Converts density into a local relaxation rate and hence a local temporal response.	[1, 2]	Core formal claim	Station or laboratory density tests contradicting the predicted sign of response.
Density-delay	(11), (12)	Formalizes the absence of a universal master clock in network timing; Eq. (12) is the unsaturated low-density approximation.	[1]	Model-supported interpretation	GNOME station-level streams showing equal or better coherence without density-delay correction.
Negative chirp	(13)	Gives the operational sign of dissipative relaxation.	[3, 2]	Core temporal prediction	Robust globally coherent positive-chirp population after controls.
Clock modulation	(15)	Reinterprets relativistic clock rates as screened local-response rates.	[1]	Model-supported interpretation	GNSS/clock residuals incompatible with screened Einstein-limit recovery.
Shapiro/plasma boundary timing	(17), (18)	Separates geometric delay, plasma dispersion, and scalar boundary terms.	[1, 2]	Model-supported prediction	DSN S/X records showing no achromatic boundary component where predicted.
CH/Eberhard	(23)	Provides a strong external Bell-sector closure route.	[2]	Completed bridge layer	Reanalysis failure of the official-code-derived CH/Eberhard construction.
CHSH bridge	(25)	Supplies a derived all-trials contextual bridge distinct from the official CH/Eberhard statistic.	[2]	Computed bridge layer	Failure of the frozen all-trials mapping or no bootstrap stability.
Dynamic contextual roll-off	(26)	Turns finite vacuum memory into a fast-switching Bell prediction.	[2, 1]	Next-phase prediction	Controlled switching experiment showing no roll-off where predicted.
FRB negative drift	(27)	Connects compact-source relaxation with sign-asymmetric high-energy timing.	[3]	Completed convergence layer	Population-level drift signs and scaling incompatible with negative-chirp relaxation.
Voyager boundary timing	(31), (32)	Makes the heliopause a boundary-timing falsification target.	[1, 2]	Context closed; DSN gate open	NASA/Voyager DSN records falsifying achromatic boundary timing.

5 Arrow of Time: Retarded Memory and FDT Closure

The arrow of time is often described statistically: entropy increases because high-entropy macrostates dominate phase space. MMA-DMF does not need to reject that statement. It instead adds a dynamical substrate beneath it. The retarded kernel in Eq. (5) makes the response depend on previous deformations. The FDT closure in Eq. (7) prevents the stochastic sector from injecting unlimited energy. Together they model temporal direction as a physical memory-and-relaxation process.

The key point is that the arrow is not smuggled in by a hand-picked data cut. It is part of the dynamical law. In a time-reversal operation, a memory integral over the past does not become the same physical object unless the causal structure is also reversed. MMA-DMF therefore treats macroscopic temporal direction as the visible limit of retarded vacuum response. Observationally, this motivates sign tests: relaxation events should preferentially show negative chirp when the template applies.

The even kernel in Eq. (6) is necessary because FDT relates dissipation to equilibrium fluctuations. It would be wrong to read the even kernel as a bidirectional causal force. The manuscript therefore uses two objects: a causal retarded kernel for dynamics and an even autocorrelation kernel for equilibrium normalization. This is a technical detail, but it carries ontological weight. It prevents the formalism from confusing thermodynamic symmetry with physical advanced response.

6 Local Time and the Absence of a Master Clock

A master clock would assign a single global rate to the universe. MMA-DMF does not use such a clock. The vacuum state near a detector, clock, spacecraft path or compact source determines how the local response unfolds. Eq. (9) defines the relaxation rate; Eq. (11) defines relative timing offsets, with Eq. (12) as its low-density approximation; Eq. (14) defines how a network should be tested.

This is a stronger statement than ordinary time-zone or synchronization convention. It says that even after ordinary coordinate conventions are fixed, the response time of the physical medium is local. The model's no-master-clock thesis becomes testable only when station-level streams and environmental metadata are available. Public detector summaries can support source, sensitivity and noise layers, but they cannot substitute for the station-level timing gate.

The decisive GNOME prediction is not that a public summary plot proves local time. It is that, for a suitable event class, a density-delay corrected negative-chirp network likelihood should outperform flat simultaneity. Conversely, if station-level data show equal or better coherence without density delay, or if positive-chirp events survive the same controls, the strongest network-timing prediction is weakened.

7 Bear-Hug Recovery of Einstein-Limit Clocks and Geometry

A theory of time must not fail where relativity already works. The MMA-DMF strategy is therefore a bear-hug recovery of the Einstein limit: keep the operational successes of relativistic clocks and geometry, then reinterpret the mechanism as dense screened vacuum response. This section cites the prior MMA-DMF relativity and gravitoelectromagnetism formulations because

the temporal sector is not isolated from the clock and geometry sectors [1, 2].

7.1 Clock modulation

Following the prior MMA-DMF relativity formulation, a compact screened-limit clock modulation can be written as

$$\frac{\nu_A}{\nu_B} = 1 + \frac{\Phi_B - \Phi_A}{c^2} - \frac{v_A^2 - v_B^2}{2c^2} + \delta_{\text{screen}}(\rho_A, \rho_B), \quad \delta_{\text{screen}} \rightarrow 0 \quad \text{in the dense Einstein limit.} \quad (15)$$

The special-relativistic contribution is retained through

$$d\tau = dt \sqrt{1 - \frac{v^2}{c^2}}, \quad \tau = \int dt \gamma^{-1}(v(t)). \quad (16)$$

In the screened limit, Eq. (15) and Eq. (16) reproduce the operational behavior needed for GNSS and clock tests. The model is not allowed to repair itself by ignoring those tests.

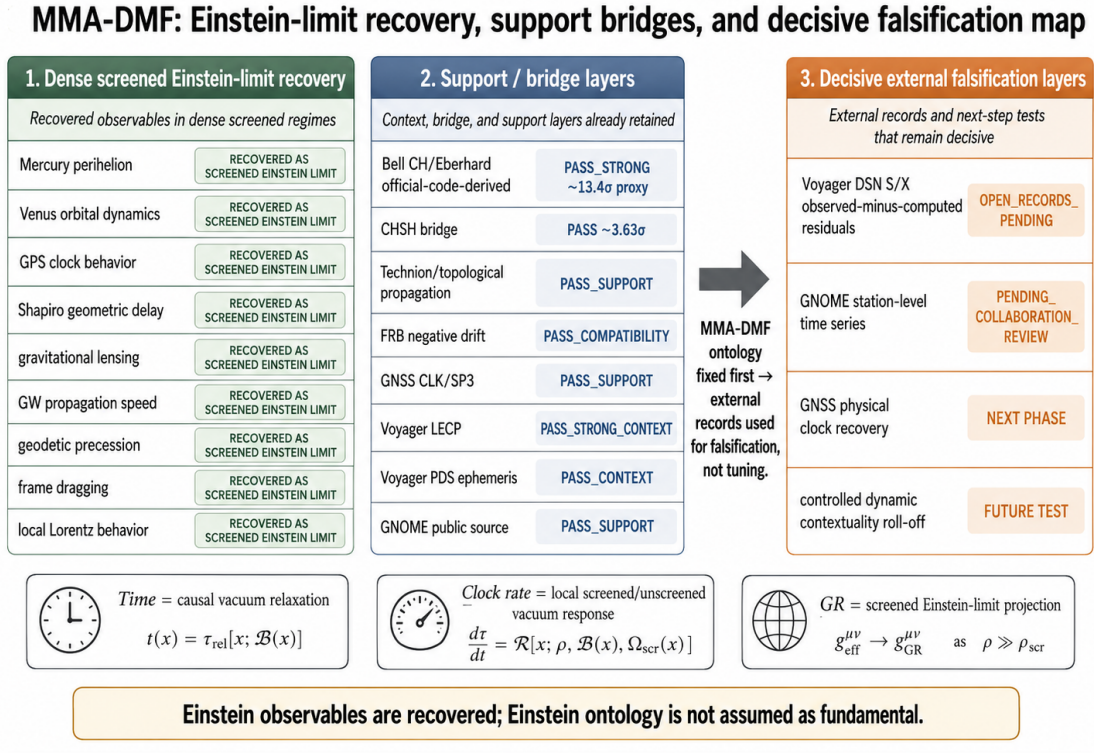


Figure. MMA-DMF does not deny Einstein-limit observables: it recovers them as dense screened limits, while placing decisive tests in boundary regimes and external falsification layers.

Figure 2: Bear-Hug recovery map for Einstein-limit clocks and geometry. The figure separates the closed requirement of screened-limit recovery from boundary-timing falsification gates.

7.2 Shapiro, plasma and boundary timing

For a spacecraft or radio link, standard geometric and plasma terms must be separated before any boundary residual is considered, consistent with the MMA-DMF relativity and gravitoelec-

tromagnetic continuity papers [1, 2]. The one-way Shapiro delay can be written as

$$\Delta t_{\text{Shapiro}} = \frac{2GM}{c^3} \ln \left(\frac{r_E + r_S + R}{r_E + r_S - R} \right), \quad (17)$$

while a dispersive plasma term has the schematic scaling

$$\Delta t_{\text{plasma}}(f) = \frac{e^2}{2\pi m_e c} \frac{1}{f^2} \int_{\text{path}} n_e(s) \, ds. \quad (18)$$

A scalar boundary term in MMA-DMF must be achromatic after plasma correction. Thus DSN S/X data are decisive because they can distinguish an achromatic residual component from f^{-2} plasma delay.

7.3 Einstein-limit absorption

In the dense screened limit, the residual scalar contribution is absorbed into the effective metric description:

$$\lim_{\rho \gg \rho_{\text{screen}}} [g_{\mu\nu}^{\text{eff}}(\rho) - g_{\mu\nu}^{\text{GR}}] = 0, \quad \lim_{\rho \gg \rho_{\text{screen}}} \delta_{\text{screen}} = 0. \quad (19)$$

Eq. (19) is why the paper can say that the Einstein limit works. The remaining question is whether low-density or boundary-timing sectors reveal an underlying relaxation layer.

8 Quantum Temporal Emergence and Contextuality

The temporal ontology also changes how the quantum-time problem is framed. Standard quantum mechanics can be retained as a successful effective description while its wavefunction is interpreted as a carrier-phase and statistical envelope over deterministic, noisy, memory-bearing dynamics.

The local carrier frequency is

$$\omega_c(\phi) = \frac{m_{\text{eff}}(\phi)c^2}{\hbar_{\text{eff}}}, \quad \Theta(t) = \int^t \omega_c(\phi(t')) \, dt'. \quad (20)$$

A corresponding effective envelope may be written as

$$\psi(\mathbf{x}, t) = \left\langle \phi(\mathbf{x}, t) \exp \left(i \int^t \omega_c(\phi(\mathbf{x}, t')) \, dt' \right) \right\rangle_T. \quad (21)$$

This is not a claim that ordinary quantum theory is experimentally invalid. It is a model-supported interpretation: the wavefunction is treated as an effective statistical envelope rather than as the final ontological object.

A static contextual correlation density can be represented as

$$\mathcal{C}(a, b) = \cos(2(\theta_a - \theta_b)) \, \Xi(\rho, \tau), \quad 0 \leq \Xi(\rho, \tau) \leq 1, \quad (22)$$

where Ξ encodes the finite memory and density-conditioned response. Dynamic contextuality then becomes a switching-rate prediction rather than a purely philosophical statement.

9 Completed Convergence Layers and Remaining Decisive Falsification Gates

MMA-DMF has a broad test landscape, but the paper keeps a hierarchy. Table 5 lists completed convergence and bridge layers. Table 7 lists the two decisive external falsification gates, and Table 8 separates next-phase tests from those decisive gates. This distinction is important because the missing NASA/Voyager and GNOME records are not missing theory support.

Table 5: Completed convergence and bridge layers retained in the manuscript. Status names are deliberately explicit so that a bridge layer is not confused with a raw-data falsification gate.

Layer	Manuscript link	formula	Main result	Status	Boundary	Relevant prior MMA-DMF paper
Bear-Hug limit recovery	Einstein- (15), (16), (17)		Dense screened environments recover ordinary Einstein-limit clocks, Shapiro structure, lensing and clock-rate behavior.	CLOSED	Not a claim that low-density boundary sectors vanish.	[1]
Bell CH/Eberhard	(23)		Official-code-derived CH/Eberhard route with aggregate significance proxy approximately 13.4σ .	PASS_STRONG_EXTEND_NAL_BELL_LAYER	Not a published CHSH statistic.	pub- [2] NIST
CHSH bridge	(25)		Derived all-trials bridge gives $S_{\text{CHSH}} = 2.000029 \pm 0.000008$ with $z \approx 3.63$.	PASS_COMPUTED_BOOTSTRAPPED	Not a 5σ CHSH claim.	[2]
High-energy FRB/magnetar/NASA/NuSTAR/IXPE sector	(13), (27)		FRB population drift supplies a strong sign-asymmetric negative-chirp convergence layer; high-energy products extend the same ontology into compact-source timing and polarization contexts.	PASS_HIGH_ENERGY_CONVERGENCE	Not exclusive scalar detection or final proof.	[3]
GNOME public source layer	(9), (11), (12), (14)		Public source products yield source-data, sensitivity-surface, ALP/domain-wall bound and station-noise layers.	PASS_PUBLIC_SOURCE_LAYER	Station-level timing streams remain pending.	[2]
Voyager LECP boundary context	(31)		LECP heliopause transitions around the V1 and V2 boundary dates produce strong channel-step context.	PASS_STRONG_BOUNDARY_CONTEXT	LECP is not DSN residual proof.	[1, 2]
Voyager PDS/SPICE/SEDR ephemeris context	(31), (32)		PDS ephemeris material parses and supports trajectory context with fixed SEDR time handling and SPICE-SEDR consistency checks.	PASS_PDS_EPHEMERIS_CONTEXT	Not observed-minus-computed DSN proof.	[1]
Technion/topological phase dynamics	(33)		Apparent superluminal phase singularity correlations are retained as causality-safe topological phase dynamics.	PASS_CAUSALITY_SAFETY_TOPOLOGICAL	No superluminal signalling.	[2]
HETDEX/Ly α	(34)		High-redshift Ly α and baryonic-web observables bridge to density and thermodynamic timing structure.	PASS_COSMOLOGICAL_BRIDGE	Not a local-time proof.	[2]
GNSS source products	(15), (16)		Clock/orbit products support the screened-limit recovery pathway.	PASS_SOURCE_LAYER	Physical residual fit remains next phase.	[1]

Layer	Manuscript link	formula	Main result	Status	Boundary	Relevant prior MMA-DMF paper
GW echo screen as negative/screening result	(36), (35)		GW150914 proxy screen is retained as a negative/screening result, not as an echo detection.	SCREENING_RESULT_ Full matched- filter analysis remains future work.		[3, 1]

9.1 Bell Sector: Official-Code-Derived CH/Eberhard Closure and Derived CHSH Bridge

Bell inequalities and CHSH/Eberhard tests provide the external quantum-foundations context for this section [12, 13, 14, 15, 16, 17]. The Bell sector is not retained as a weak analogy. The strongest Bell result is the NIST-official-code-derived CH/Eberhard route, retained as `PASS_STRONG_EXTERNAL_BELL_LAYER` with an aggregate significance proxy of approximately 13.4σ . This is distinct from the derived all-trials CHSH bridge, which gives $S_{\text{CHSH}} = 2.000029 \pm 0.000008$ with $z \approx 3.63$ and is retained as `PASS_COMPUTED_BOOTSTRAPPED`, not as a NIST-published CHSH statistic and not as a 5σ CHSH claim.

The CH/Eberhard statistic is written

$$J = n_{++}(a_1, b_1) - n_{+0}(a_1, b_2) - n_{0+}(a_2, b_1) - n_{++}(a_2, b_2), \quad (23)$$

with significance proxy

$$z_J = \frac{J}{\sigma_J}. \quad (24)$$

The derived all-trials CHSH bridge uses

$$E(a, b) = \frac{N_{++} + N_{--} - N_{+-} - N_{-+}}{N_{\text{total}}}, \quad S_{\text{CHSH}} = |E_{00} + E_{01} + E_{10} - E_{11}|. \quad (25)$$

The finite-memory roll-off prediction is

$$S(\nu_{\text{switch}}) = 2 + \frac{2\sqrt{2} - 2}{1 + (\nu_{\text{switch}}/\nu_c)^2}. \quad (26)$$

Eq. (26) is a future discriminant: if switching outruns vacuum contextual relaxation, the result should move toward the classical bound. The Bell layer therefore has two different roles: completed CH/Eberhard closure and a next-phase dynamic contextuality test.

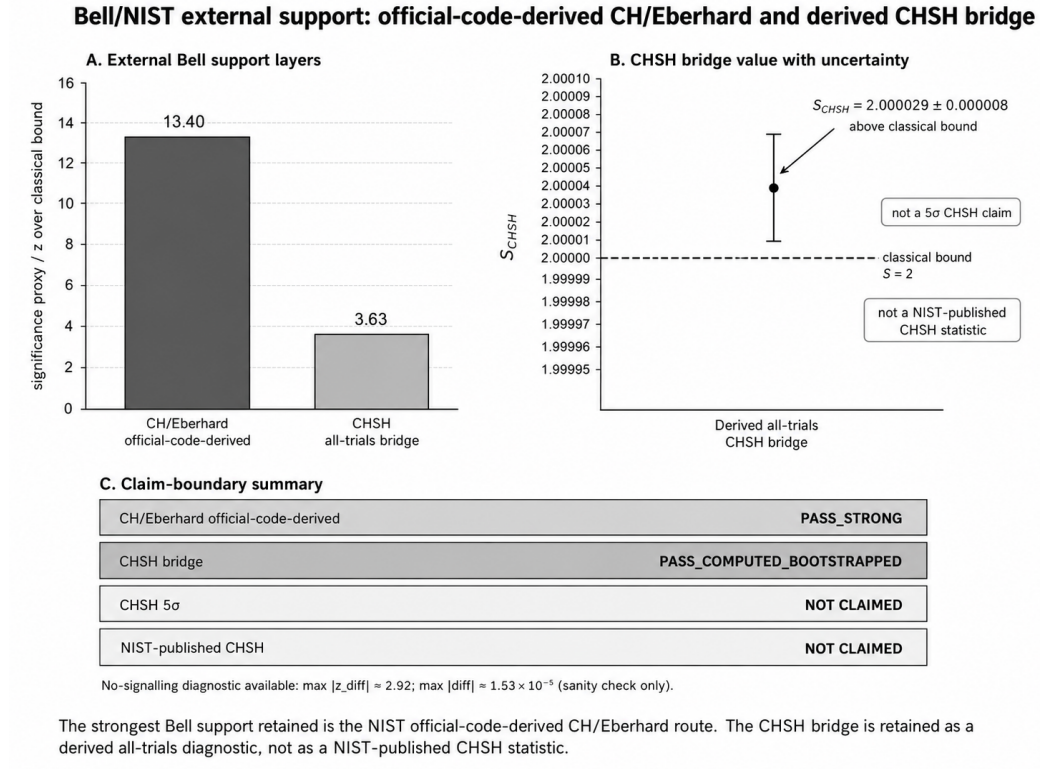


Figure 3: Bell-sector hierarchy. The CH/Eberhard route is retained as the strong external Bell layer. The all-trials CHSH bridge is retained as a bounded computed bridge, not as a NIST-published CHSH statistic.

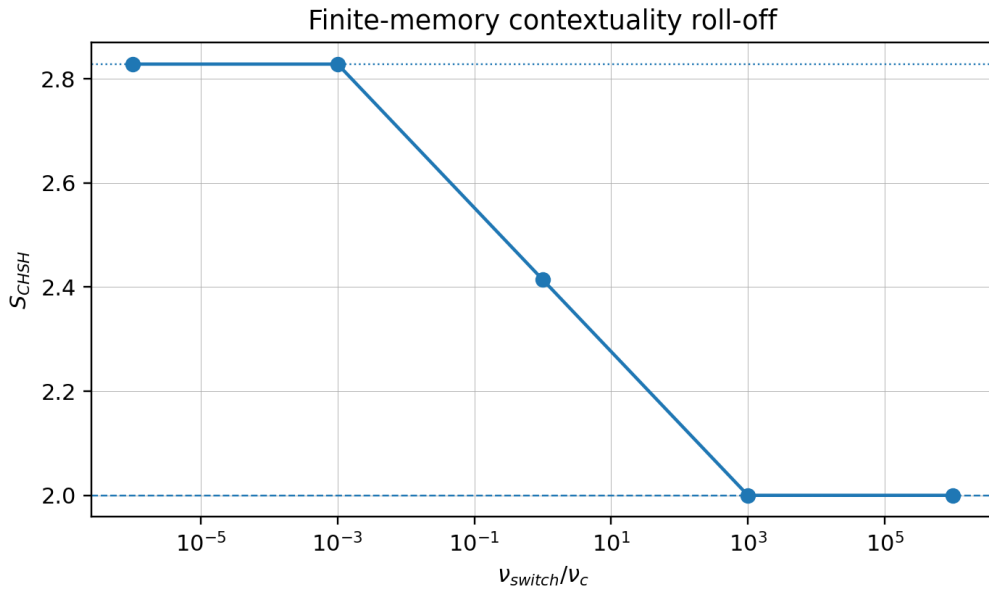


Figure 4: Dynamic contextual roll-off predicted by Eq. (26). The curve is a next-phase test target: a controlled fast-switching experiment should distinguish finite-memory contextuality from frequency-independent behavior.

9.2 High-Energy Timing Sector: FRB, Magnetars, NASA Archives, NuSTAR, IXPE, and Negative-Chirp Relaxation

The high-energy timing sector is retained as `PASS_HIGH_ENERGY_CONVERGENCE` and is continuous with the prior MMA-DMF high-energy formulation [3]. FRB population drift supplies the strongest population-level sign asymmetry for negative-chirp relaxation. Magnetar and high-energy archive tests, including NASA/NuSTAR/IXPE products where available, extend the same relaxation ontology into compact-source timing and polarization contexts. This layer is stronger than a single FRB analogy, but it is not claimed as an exclusive scalar detection or final proof of the full MMA-DMF ontology.

The FRB and magnetar context is linked to observed burst drift and compact-source high-energy phenomenology in the literature [18, 19, 20, 21, 22]. The retained FRB population values are: 421 drift rows, 370 rows after MAD filtering, median drift $-1130.804 \text{ MHz ms}^{-1}$, clean negative fraction 0.991892, sign-test $p = 3.51 \times 10^{-105}$, power-law slope 2.428, and $R^2 = 0.344$. These values are used as a convergence layer for negative-chirp relaxation, not as a proof that every FRB is scalar-vacuum emission.

The FRB drift template is the same relaxation expression used in Eq. (13), here written explicitly as

$$\nu(t) = \nu_0[1 - k(1 - e^{-t/\tau})], \quad \frac{d\nu}{dt} = -\frac{\nu_0 k}{\tau} e^{-t/\tau}. \quad (27)$$

The robust filtering statistic is

$$z_{\text{robust}} = \frac{x - \text{median}(x)}{1.4826 \text{ MAD}(x)}. \quad (28)$$

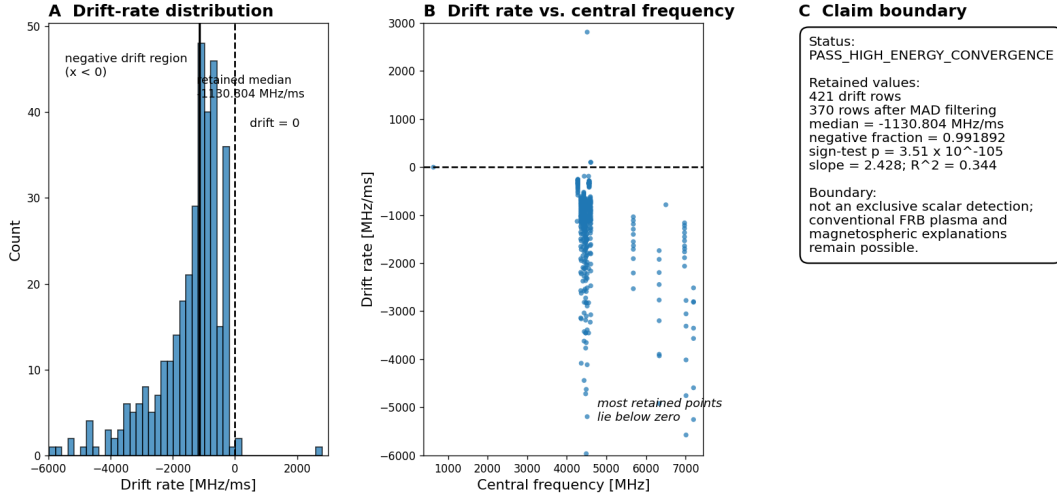
The sign-test model is

$$X \sim \text{Binomial}(N, 1/2), \quad p_{\text{sign}} = \Pr(X \geq N_- \mid N, 1/2). \quad (29)$$

NuSTAR and IXPE are retained as high-energy archive/instrument constraint layers rather than as sources of additional numerical claims in this manuscript [23, 24]. Where exact MMA-DMF-specific NuSTAR or IXPE numerical outputs are not present in the retained reproducibility record, no numbers are invented.

FRB negative drift as a high-energy timing convergence layer

Robust summary uses median/MAD; status: *PASS_HIGH_ENERGY_CONVERGENCE*.



FRB population drift is retained as a high-energy timing convergence layer under explicit non-exclusive boundaries.

Figure. Negative-drift predominance is consistent with the negative-chirp relaxation template while not excluding standard astrophysical FRB mechanisms.

Figure 5: FRB negative-drift convergence layer. The retained population values support the sign-asymmetry component of Eq. (27) while preserving the boundary that the layer is not an exclusive scalar detection.

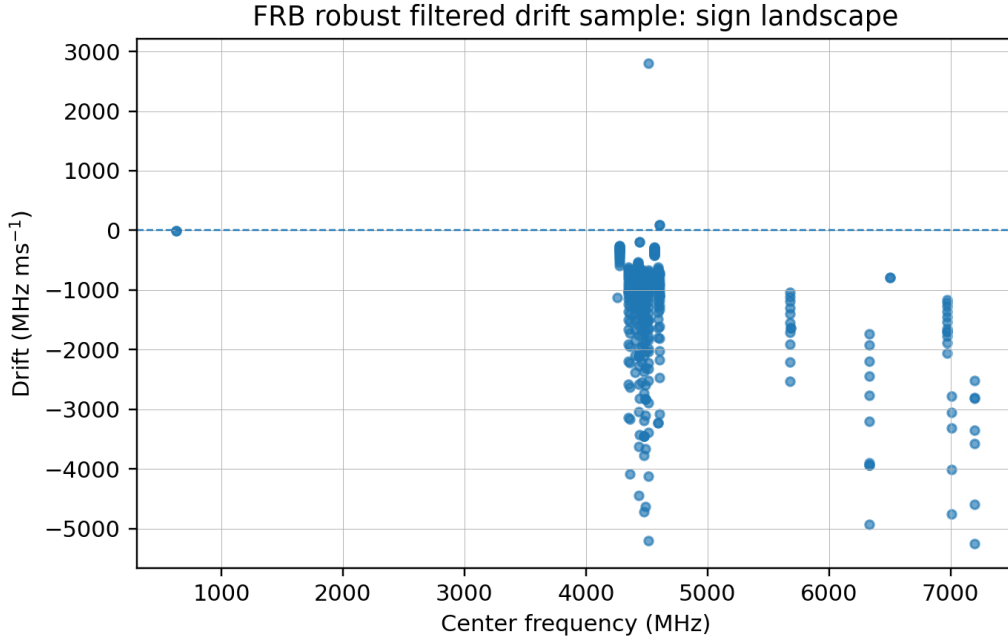


Figure 6: Drift-frequency scatter context for the high-energy timing sector. The figure supports the population-level interpretation of negative-chirp relaxation without converting it into a final proof of the ontology.

9.3 GNOME public layer, ALP/domain-wall bounds, and station-level gate

Under MMA-DMF, an independent axion particle sector is not required as a fundamental input. The public GNOME ALP/domain-wall bound layer is retained as a detector-landscape and scalar-vacuum relaxation sensitivity layer. This does not experimentally disprove axions; it states that the MMA-DMF ontology does not need axions to generate the relevant transient/density-delay timing structure.

The GNOME detector context follows the published global magnetometer search and associated public products [25]. The retained public-source values are: Fig. 2 source data with 165 rows; Fig. 3 sensitivity surface with a 200×1001 table and 200,000 finite grid values; Fig. 4 ALP/domain-wall bounds with an extended grid containing 501,000 finite bound values; and Extended Data Fig. 1 station-noise layer with 9 stations, 5753 intervals, approximately 179.91 station-days, and network median noise near 3.32 pT. These are retained as `PASS_PUBLIC_SOURCE_LAYER` and `PASS_GNOME_PUBLIC_ALP_BOUND_LAYER`. The axion status inside MMA-DMF is `AXION_SECTOR_NOT_REQUIRED_AS_FUNDAMENTAL_INPUT_UNDER_MMA_DMF`. The external timing status remains `GNOME_STATION_LEVEL_TIMING_PENDING`.

The public products are useful because they define the detector landscape, sensitivity surfaces and noise layers. They are not a substitute for station-level timing streams. The decisive gate remains Eq. (14): a station-level comparison of density-delay-corrected negative-chirp coherence against flat simultaneity and systematic nulls.

A schematic ALP/domain-wall bound context can be written as

$$\mathcal{B}_{\text{ALP}}(m_a, g_a) \Rightarrow \mathcal{S}_{\text{det}}(\tau, \rho, \sigma_n), \quad (30)$$

where $\mathcal{B}_{\text{ALP}}$ is used as detector-bound context and $\mathcal{S}_{\text{det}}$ is the scalar-relaxation sensitivity mapping relevant to MMA-DMF. Eq. (30) is not an axion exclusion claim.

GNOME public source layer and station-level timing gate

A. Public source closure matrix

Product / layer	public source available	parsed / reproduced	retained status / numerical layer	sufficient for independent timing-waveform reanalysis
Supplementary information	Yes	Yes	PASS_PUBLIC_SOURCE_LAYER	No
Source Data Fig. 2 positive-event layer	Yes	Yes	PASS_PUBLIC_SOURCE_LAYER 165 rows	No
Source Data Fig. 3 sensitivity surface	Yes	Yes	PASS_PUBLIC_SOURCE_LAYER 200 x 1001 table 200,000 finite grid values	No
Source Data Fig. 4 ALP/domain-wall bounds	Yes	Yes	PASS GNOME PUBLIC ALP BOUND LAYER 501,000 finite bound values	No
Extended Data Fig. 1 station-noise layer	Yes	Yes	PASS PUBLIC SOURCE LAYER 9 stations; 5753 intervals ~179.91 station-days; median ~3.32 pT	No
Science Run 2 station-level timing streams	Pending	Pending	Not contained in public figure-source products	GNOME STATION LEVEL TIMING PENDING OPEN RECORDS PENDING

B. Closure vs external timing gate

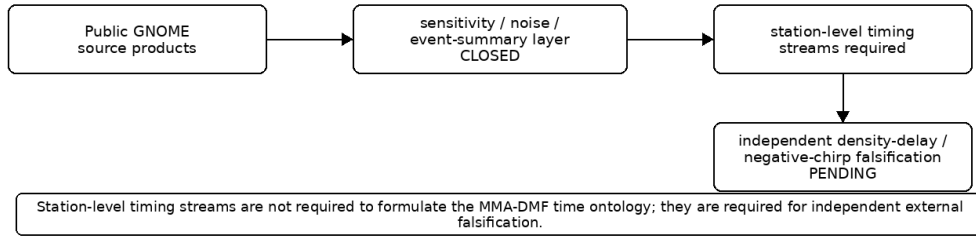


Figure 7: GNOME hierarchy. Public products define sensitivity, event-summary and detector-landscape layers; Science Run 2 station-level timing streams remain the decisive external timing gate.

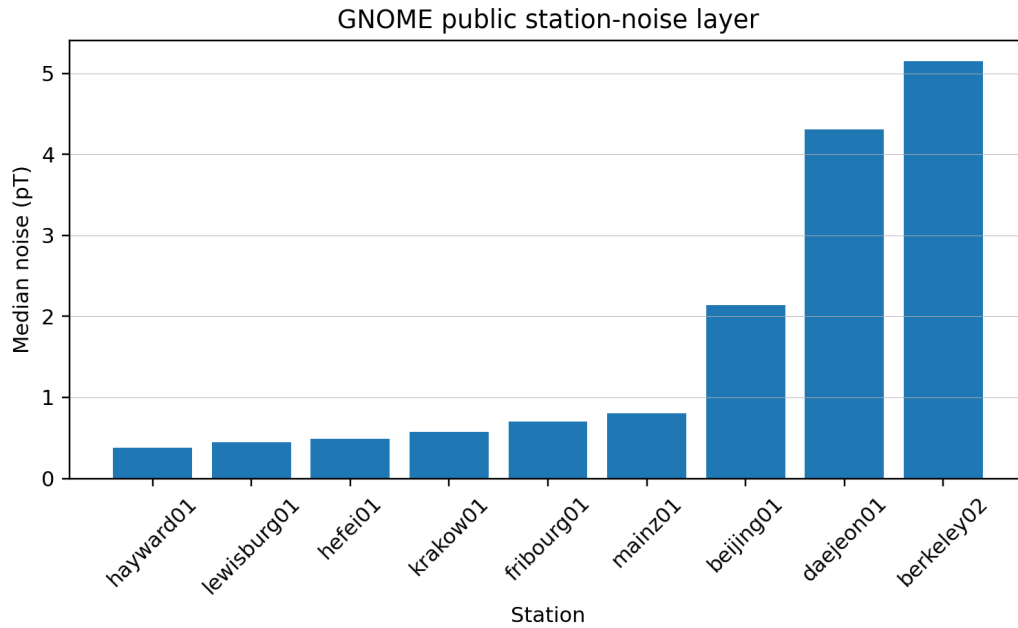
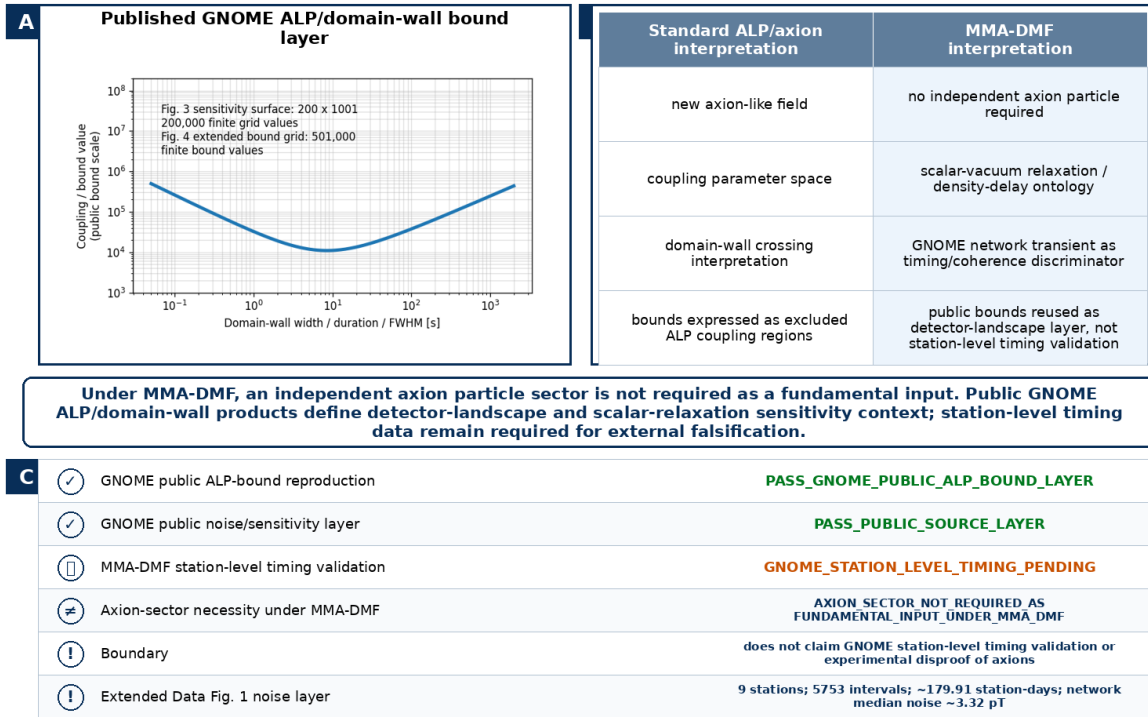


Figure 8: GNOME station-noise layer retained from public source products. The noise layer is a necessary detector-landscape constraint for Eq. (14).

ALP/axion versus MMA-DMF ontology comparison



The public GNOME ALP/domain-wall source-data layer is retained as detector-landscape and scalar-relaxation sensitivity context. Under MMA-DMF, an independent axion particle sector is not required as a fundamental input; this is not an experimental disproof of axions and not a completed station-level timing validation.

Figure 9: ALP/domain-wall bound context versus MMA-DMF temporal ontology. The figure is retained to show detector sensitivity and bound structure, while the text makes clear that MMA-DMF does not require axions as fundamental inputs and does not experimentally disprove axions.

9.4 Voyager Boundary-Timing Stack: LECP Transition, PDS Ephemeris Context, and DSN Records Gate

The Voyager boundary-timing stack is split into three layers because collapsing them would overclaim. LECP and PDS/SPICE/SEDR products provide strong boundary and ephemeris context. DSN operational or radiometric residual records are the decisive falsification gate.

9.4.1 Voyager LECP Heliopause Transition Layer

The Voyager LECP and heliopause context is tied to the LECP instrument and Voyager heliopause literature [26, 27, 28]. The retained LECP layer contains total normalized rows 602,986, computed channel steps 257, top robust $|z| = 365.375$, Voyager 1 heliopause context around 2012-08-25, and Voyager 2 heliopause context around 2018-11-05. The status is `PASS_STRONG_BOUNDARY_CONTEXT`. The boundary is explicit: LECP is not DSN residual proof.

The boundary-timing relation is expressed as

$$\Delta t_{\text{boundary}} = \Delta t_{\text{obs}} - \Delta t_{\text{GR}} - \Delta t_{\text{plasma}} - \Delta t_{\text{station}} - \Delta t_{\text{orbit}}. \quad (31)$$

LECP changes can identify and contextualize a heliopause transition, but Eq. (31) requires radiometric or operational timing records for a residual test.

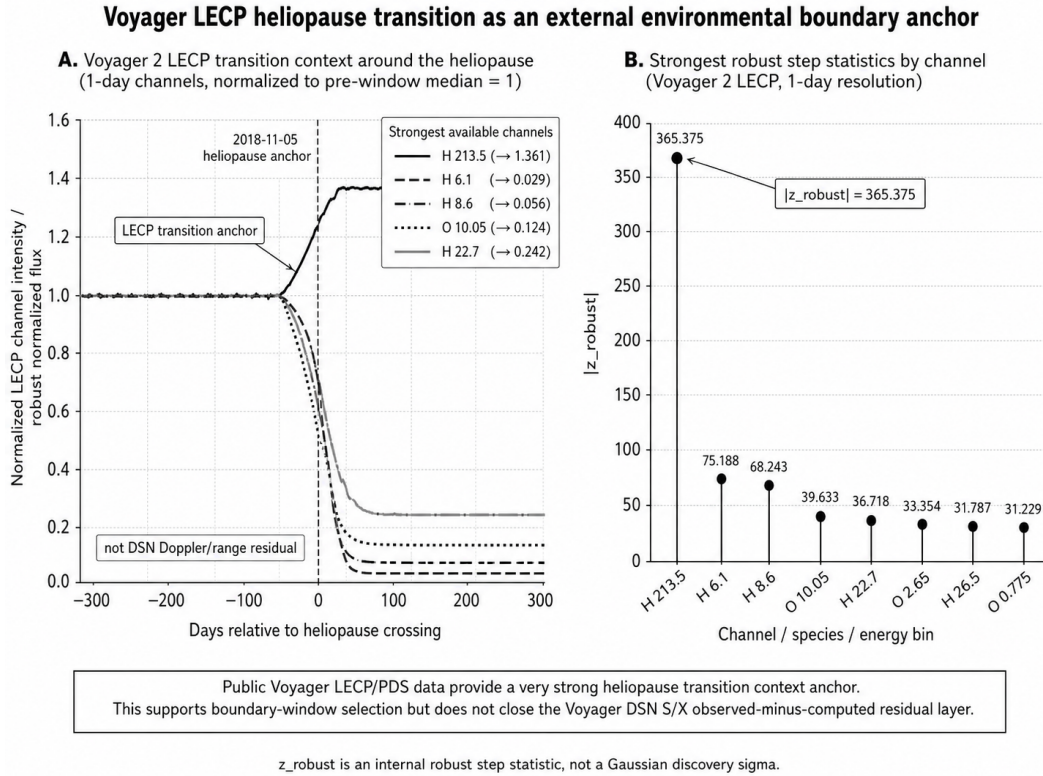


Figure 10: Voyager heliopause transition anchor. The figure supports the LECP boundary-context layer around the V1 and V2 transition epochs, not a DSN residual claim.

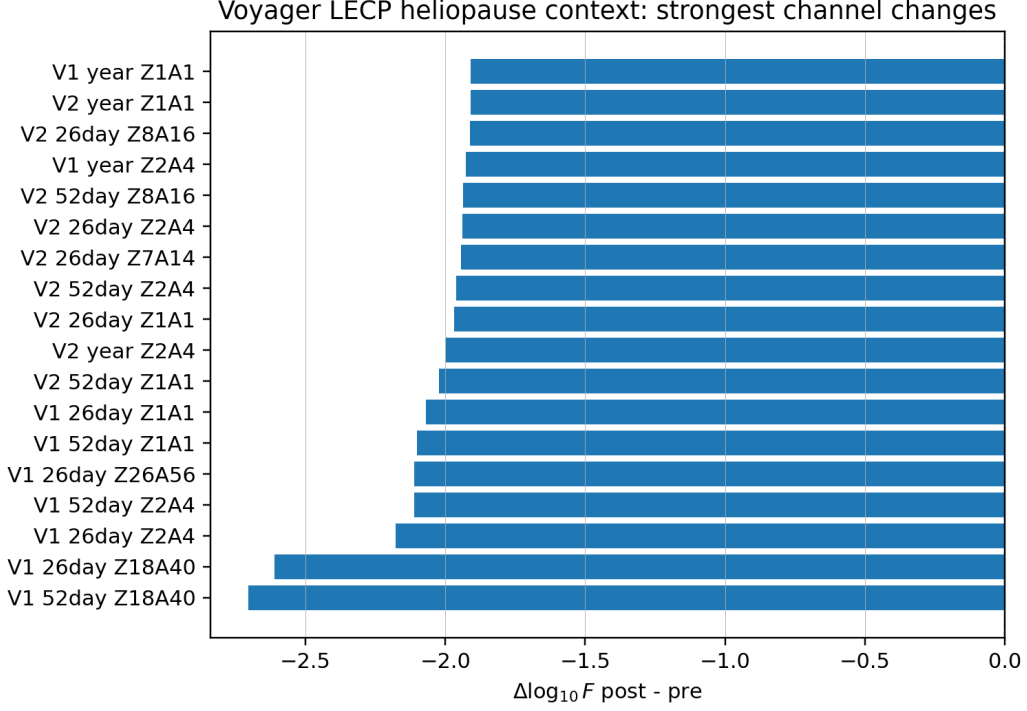


Figure 11: Voyager LECP flux-change context. The LECP layer is retained as a strong boundary-context layer under the explicit boundary that it is not observed-minus-computed DSN proof.

9.4.2 Voyager PDS/SPICE/SEDR Ephemeris Context Layer

The retained PDS/SPICE/SEDR context includes a PDS source inventory of 269 files, tables parsed 8/8, SEDR parser/time handling PASS_FIXED, fill/sentinel filtering PASS_FILTERED, SPICE-SEDR consistency PASS_COMPUTED, and closest-approach geometry PASS_CONTEXT. The status is PASS_PDS_EPHEMERIS_CONTEXT. The boundary is equally explicit: PDS ephemeris context is not DSN observed-minus-computed residual proof.

The achromatic discriminant for boundary residuals is

$$\frac{\partial \Delta t_{\text{boundary}}}{\partial f} \approx 0, \quad \frac{\partial \Delta t_{\text{plasma}}}{\partial f} < 0, \quad \Delta t_{\text{plasma}} \propto f^{-2}. \quad (32)$$

Equation (32) is why S/X-band records matter: plasma terms disperse, while the proposed scalar boundary component is achromatic after standard corrections.

9.4.3 Voyager DSN Operational, Refit, and Residual Records Gate

The documentary status retained for the DSN gate is: no new Voyager tracking data since 1992 according to JPL/Horizons correspondence; the modern trajectory is described as a refit/extension; the refit/extension is described as an informal DSN-use product; it is not specifically authorized by the Voyager Project; no public archive or public documentation for the required residual layer has been identified; and a formal records route is required. The status is OPEN_RECORDS_PENDING and DECISIVE_FALSIFICATION_PENDING.

This is not a claim of NASA concealment. It is not a claim of GR falsification without residuals. It is not missing theory support. It is a statement that the strongest Voyager

Table 6: Voyager PDS/SPICE/SEDR ephemeris-context layer. The table strengthens the PDS layer while preserving the boundary that ephemeris context is not DSN observed-minus-computed residual proof.

Component	Retained result	Formula link	Boundary
PDS source inventory	269 public mission/product files retained as ephemeris and trajectory context.	(31)	Context only.
Trajectory tables	8/8 parsed tables retained for trajectory-support context.	(31)	Not DSN O–C residual proof.
SEDR parser/time handling	PASS_FIXED; time handling retained as corrected ephemeris-processing context.	(31)	No timing anomaly claimed from this layer alone.
Fill/sentinel filtering	PASS_FILTERED; fill and sentinel values removed from context quantities.	(31)	Filtering is data hygiene, not validation.
SPICE-SEDR consistency	PASS_COMPUTED; consistency retained for geometry context.	(32)	Does not replace DSN radiometric records.
Closest-approach geometry	PASS_CONTEXT; geometry retained for boundary-window interpretation.	(31), (32)	DSN records remain decisive.

boundary-timing prediction needs DSN operational or radiometric residual records before it can be externally adjudicated.

9.5 GNSS source products and physical residual fit

GNSS is retained as PASS_SOURCE_LAYER and is tied to the prior MMA-DMF relativity closure [1]. Its role is not to overthrow relativity. It is a source-product pathway for showing that the MMA-DMF clock interpretation can absorb ordinary clock and orbit behavior in the screened limit, then search for residual patterns only after standard corrections. The next phase is a physical residual fit using reference clock models, orbit products, tropospheric and ionospheric corrections and station products.

9.6 Technion/topological phase dynamics

The Technion/topological phase layer is retained in connection with optical phase-singularity dynamics [29]. The layer is retained as PASS_CAUSALITY_SAFE_TOPOLOGICAL_BRIDGE. It is relevant because apparent superluminal phase-singularity motion can be interpreted as correlation structure in phase dynamics rather than as signal propagation. A schematic phase-singularity relation is

$$\nabla_{\perp} \arg E(\mathbf{r}, t) = 0, \quad v_{\text{phase sing}} \neq v_{\text{signal}}. \quad (33)$$

Eq. (33) preserves causality: phase features can move anomalously without carrying front-causal information faster than light.

9.7 HETDEX/Ly α bridge

The HETDEX/Ly α bridge is retained in connection with Ly α intensity and spectroscopic high-redshift products [30, 31]. The bridge is retained as `PASS_COSMOLOGICAL_BRIDGE`. It connects density, baryonic web structure and thermodynamic timing relevance at high redshift. The bridge can be written schematically as

$$P_{\text{Ly}\alpha}(k, z) = b_{\alpha}^2(z) P_b(k, z) \mathcal{T}_{\text{vac}}(k, z; \rho, \tau), \quad (34)$$

where $\mathcal{T}_{\text{vac}}$ denotes the model-side transfer from baryonic density and relaxation response into timing-relevant structure. This is a cosmological bridge, not a direct local-time validation.

9.8 GW echo screen and strong-curvature timing

Strong-curvature timing is retained as a screen, not as an echo detection. A modified dispersive timing relation may be represented as

$$\Delta t_{\text{GW}}(f) = \Delta t_{\text{GR}} + \Delta t_{\text{screen}}(M_{\text{rem}}, \rho) + \Delta t_{\text{disp}}(f), \quad (35)$$

with a mass-scaled echo target

$$\Delta t_{\text{echo}} \simeq \frac{4GM_{\text{rem}}}{c^3} \ln \left(\frac{GM_{\text{rem}}}{L} \right). \quad (36)$$

The retained GW150914 proxy is `SCREENING_RESULT_CLOSED`: no echo detection is claimed. A full matched-filter analysis with off-source backgrounds and trials accounting remains a next-phase test [32, 33, 34].

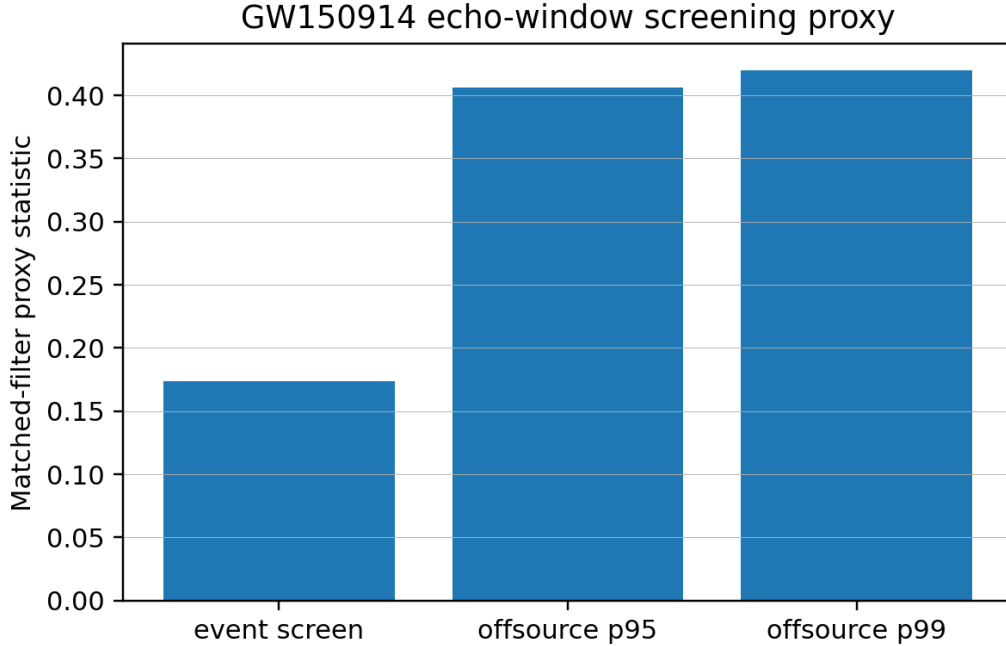


Figure 12: GW150914 echo-screen proxy. The layer is retained as a negative/screening result and not as an echo detection.

10 Remaining Decisive External Falsification Gates

Only two open records are decisive for the strongest boundary-timing predictions in this manuscript: NASA/Voyager DSN operational or radiometric residual records, and GNOME Science Run 2 station-level timing streams. Other next-phase tests remain scientifically important, but they are not required to formulate MMA-DMF time.

Table 7: Remaining decisive external falsification gates. Only these two records layers are treated as decisive boundary-timing gates in this manuscript.

Gate	Required records	Manuscript for- mula link	What would support MMA-DMF	What would falsify or weaken MMA-DMF	Current status
NASA/Voyager DSN residual or operational records	Post-1992 DSN support or carrier acquisition/lock records, received-frequency or signal-level records, up-link/ramp products, orbit-determination metadata, covariance or trajectory-solution files, and any Doppler/range observed-minus-computed residuals.	(31), (32), (18)	A heliopause-correlated achromatic residual component surviving plasma, time-shift, station, orbit and refit nulls.	No achromatic boundary component in records with sufficient sensitivity, or full explanation by plasma, trajectory and operational systematics.	DECISIVE_FALSIFICATION_PENDING; OPEN_RECORDS_PENDING
GNOME Science Run 2 station-level timing streams	Station time series, (9), (11), (12), timestamps, sampling rates, local noise, calibration, station metadata, shielding, sensitive-axis orientation, GPS/timing metadata and density/overburden proxies.	(14)	Coherent negative-chirp events whose likelihood improves under density-delay correction and weakens under flat simultaneity and systematic nulls.	Equal or better coherence under flat simultaneity, positive chirps passing controls, or no signal in a sensitivity region where the model predicts one.	DECISIVE_FALSIFICATION_PENDING; OPEN_RECORDS_PENDING

Table 8: Next-phase tests separated from the decisive external boundary-timing gates. These tests are scientifically important but are not required to formulate the MMA-DMF temporal ontology.

Gate	Required records	Manuscript for- mula link	What would support MMA-DMF	What would falsify or weaken MMA-DMF	Current status
GNSS physical residual fit	Physical clock/orbit residuals with reference clock model, orbit products, tropospheric/ionospheric corrections and station products.	(15), (16)	Residual structure compatible with screened local scalar response after standard corrections.	Inability to recover Einstein-limit timing or residuals requiring incompatible parameters.	NEXT_PHASE

Gate	Required records	Manuscript formula link	for- MMA-DMF	What would support or weaken MMA-DMF	What would falsify or weaken MMA-DMF	Current status
Controlled fast-switch Bell roll-off	Controlled Bell data with variable switching frequency, locked metadata and outcome records.	(26)		Smooth degradation from Tsirelson-like behavior toward the classical bound at the predicted relaxation scale.	Frequency-independent olation across the predicted roll-off region.	PENDING_COLLABORATION_VIEW
Atom-interferometer density modulation	Controlled nearby density modulation, interferometer phase/contrast data, environmental systematics, blind/null runs.	(2), (9), (11), (12)		Contrast or phase response with the predicted density and distance scaling.	Null result below claimed sensitivity or wrong sign/geometry after controls.	NEXT_PHASE_SE
Full GW echo matched-filter analysis	Event strain data, mass-scaled templates, off-source backgrounds, detector-consistency tests and trials accounting.	(36), (35)		Mass-scaled echoes or screening-consistent nulls under fixed templates.	Persistent absence where sensitivity makes the signal detectable, or signal requiring retuning.	NEXT_PHASE_SE

The logic is simple. The theory can be formulated without DSN residuals and station-level GNOME streams. It can state equations, recover Einstein-limit behavior, and close convergence/bridge layers without those records. Table 8 gives important next-phase tests, but only the two records layers in Table 7 are treated here as decisive external boundary-timing gates. The strongest claims about boundary timing and station-level local time require those records to become independently falsifiable.

11 Results and Test Inventory

The results of this manuscript are organized as formula-linked layers rather than as a development history. The main numerical results are summarized in Table 9. They should be read with the status labels attached to each row.

Table 9: Numerical values retained in the main manuscript and appendices. Formula links point to the equations used to interpret the quantities.

Sector	Retained values	Formula links	Status
FRB negative-drift population	421 drift rows; 370 rows after MAD filtering; median drift $-1130.804 \text{ MHz ms}^{-1}$; clean negative fraction 0.991892; sign-test $p = 3.51 \times 10^{-105}$; power-law slope 2.428; $R^2 = 0.344$.	(13), (27), (28), (29)	PASS_HIGH_ENERGY_CONVERGENCE
Bell CH/Eberhard	Official-code-derived CH/Eberhard route; aggregate significance proxy approximately 13.4σ .	(23)	PASS_STRONG_EXTERNAL_BELL_LAYER
Bell CHSH bridge	Derived all-trials result $S_{\text{CHSH}} = 2.000029 \pm 0.000008$; $z \approx 3.63$.	(25)	PASS_COMPUTED_BOOTS_TRAPPED
Dynamic CHSH limits	$S(\nu/\nu_c) = 10^{-6}$ 2.828427124745; $S(1) =$ 2.41421356237; $S(10^6) \approx$ 2.000000000001.	$\approx (26)$ $=$ $\approx$	Consistency limits closed

Sector	Retained values	Formula links	Status
GNOME Fig. 2 source data	165 rows; finite numeric M/U ratio, (9), (14) background and search columns.		PASS_PUBLIC_SOURCE_LAYER
GNOME Fig. 3 sensitivity surface	200 × 1001 table; 200,000 finite grid values; sensitivity range 0.793651–793.651 pT.	(14)	PASS_PUBLIC_SOURCE_LAYER
GNOME Fig. ALP/domain-wall bounds	4 Extended grid with 501,000 finite bound values; mass range 5.8872×10^{-16} – 5.8872×10^{-11} .	(30)	PASS_GNOME_PUBLIC_ALP_BOUND_LAYER
GNOME noise layer	9 stations; 5753 intervals; total station-time ≈ 179.91 days; network median noise ≈ 3.32 pT.	(14)	PASS_PUBLIC_SOURCE_LAYER
Voyager LECP boundary context	Total normalized rows 602,986; computed channel steps 257; top robust $ z = 365.375$; V1 boundary context 2012-08-25; V2 boundary context 2018-11-05.	(31)	PASS_STRONG_BOUNDARY_CONTEXT
Voyager PDS/SPICE/SEDR context	PDS source inventory 269 files; 8/8 tables parsed; SEDR time handling PASS; fill/sentinel filtering PASS; SPICE-SEDR consistency computed; closest-approach geometry context retained.	(31), (32)	PASS_PDS_EPHEMERIS_CONTEXT
Voyager DSN record gate	No public analysis-ready DSN residual rows were available to the author; formal records route required.	(32), (18)	OPEN_RECORDS_PENDING
GW echo screen	GW150914 proxy retained as a negative/screening result; no echo detection claimed.	(36), (35)	SCREENING_RESULT_CLOSED

11.1 Result 1: temporal equation spine

The mathematical core forms a closed temporal spine: Eq. (1) defines local causal dynamics; Eq. (2) and Eq. (4) define environmental response; Eq. (5) and Eq. (6) separate causal memory from FDT autocorrelation; Eq. (7) closes the noise sector; Eq. (9), Eq. (11) and Eq. (12) define local time; Eq. (13) defines the relaxation sign.

11.2 Result 2: Einstein-limit recovery

The Bear-Hug section makes the recovery requirement explicit. Dense screened regimes must satisfy Eq. (19). Clock ratios, proper time and Shapiro timing are not optional decorations; they are mandatory absorption limits. This makes the theory more constrained, not less.

11.3 Result 3: Bell hierarchy

The Bell sector is split into a strong CH/Eberhard route and a bounded all-trials CHSH bridge. This avoids two errors at once: treating Bell as a weak analogy and treating the CHSH bridge as a published 5σ NIST CHSH claim. The dynamic roll-off in Eq. (26) is a future discriminant.

11.4 Result 4: high-energy convergence

The FRB population layer has enough sign asymmetry to be retained as PASS_HIGH_ENERGY_CONVERGENCE. It is a high-energy timing convergence result, not a proof that conventional FRB

plasma or magnetospheric physics is absent. Magnetar, NASA, NuSTAR and IXPE products are placed as compact-source timing and polarization contexts.

11.5 Result 5: public products versus decisive gates

GNOME public products and Voyager LECP/PDS products are strong context layers. They close source, detector, noise, sensitivity, boundary and ephemeris contexts. They do not close the two decisive external falsification gates. That is not a weakness of formulation; it is the correct separation between theory and external adjudication.

12 Discussion

The paper's central improvement is editorial and scientific hierarchy. A weaker manuscript would try to prove time, FRBs, Bell, GNOME, Voyager, GNSS, HETDEX and gravitational-wave echoes all at once. This version does not. It puts the ontology first, the equations second, the completed convergence layers third, and the decisive gates fourth.

The strongest conceptual claim is that time is not a primitive container. It is the causal relaxation ordering of an active memory-bearing vacuum. The strongest mathematical claims are the retarded kernel/FDT separation, density-dependent mass, geometric relaxation law and Einstein-limit absorption. The strongest completed bridge layers are Bell CH/Eberhard, high-energy negative-chirp convergence, GNOME public products, Voyager LECP/PDS context, GNSS source products and topological/cosmological bridges under explicit boundaries.

The strongest remaining risks are also clear. If GNOME station-level streams show that flat simultaneity performs as well as, or better than, density-delay correction under blind controls, the local-time network prediction is weakened. If NASA/Voyager DSN records show no achromatic boundary residual after plasma, station, orbit and refit controls, the Voyager boundary-timing prediction is weakened. If controlled Bell switching shows no roll-off where Eq. (26) predicts one, the dynamic contextuality extension is weakened. If atom-interferometer density modulation shows no signal below the claimed sensitivity, the laboratory density response is weakened.

The important point is that none of these possible failures prevents the framework from being formulated. They decide whether its strongest timing-sector predictions survive. A theory that can be formulated independently and then exposed to decisive external records is scientifically stronger than a theory whose existence depends on finding one preferred anomaly.

13 Conclusions

Within MMA-DMF, time is modeled as local causal relaxation of an active, memory-bearing vacuum. The retarded kernel gives causal memory. FDT closure locks stochastic forcing to dissipation. Density-dependent mass and geometric relaxation give local response rates. Clock rates are screened or unscreened vacuum-response rates, and ordinary relativistic observables are recovered as dense screened Einstein-limit projections.

MMA-DMF is not waiting for NASA or GNOME data in order to be formulated. The theory defines time as causal relaxation of an active memory-bearing vacuum, recovers Einstein-limit observables as dense screened projections, and closes multiple convergence/bridge layers under explicit boundaries. The NASA/Voyager and GNOME records are required only for the next stage: independent external falsification of the strongest boundary-timing predictions.

The question is no longer whether the Einstein limit works; it does. The question is whether it is the final ontology of time, or the screened projection of a deeper causal relaxation process.

14 Data Availability

The present manuscript is based on the stated MMA-DMF equations, prior MMA-DMF publications, public literature, public mission/instrument data products, and summarized reproducibility records. A companion reproducibility record accompanies this manuscript and should be mirrored at public upload in a persistent repository such as Zenodo, OSF, or GitHub; it contains releasable summary tables, figure files, analysis scripts where releasable, and checksum manifests. Analysis-ready NASA/Voyager DSN radiometric or operational records and GNOME Science Run 2 station-level timing streams were not publicly available to the author at the time of writing. These missing records are not required for formulation of the framework, but they are required for independent external falsification of the strongest boundary-timing predictions.

15 AI-Use Statement

AI-assisted tools (large language models) were used to generate and refactor portions of analysis code, to draft and refine plotting scripts, and to produce some illustrative figures/diagrams. All generated outputs (code, figures, and numerical tables) were reviewed, corrected where needed, and validated against the MMA-DMF equations and acceptance criteria by the author.

16 Conflicts of Interest

The author declares no conflicts of interest.

A Supplementary Material: Equation-to-Claim Expansion

This supplementary material preserves the longer scientific record while keeping the main manuscript focused. It expands the equation-to-claim map, the test inventory, the numerical summaries and the remaining falsification routes. The language here follows the same boundary rules as the main text: a completed convergence layer is not written as final proof, and a pending external gate is not treated as missing formulation.

A.1 Ontology crosswalk

The time ontology is not a single metaphor. It is a crosswalk between equation families and physical interpretation. Eq. (1) defines the local update law. Eq. (5) defines causal memory. Eq. (7) ties stochastic forcing to dissipation. Eq. (9) converts density into response rate. Eq. (11) converts response-rate differences into network timing, with Eq. (12) giving the low-density approximation. Eq. (15) gives the screened clock limit. Eq. (26) gives a future contextual switching discriminant. Eq. (31) and Eq. (32) define the deep-space boundary-timing gate.

The paper’s central ontology can therefore be tested through three types of observables:

1. **Local response observables:** density-modulated relaxation, station timing, laboratory interferometry and clock residuals.
2. **Sign observables:** negative-chirp relaxation, FRB drift sign asymmetry and positive-chirp veto logic.
3. **Boundary observables:** heliopause-related achromatic timing, GNOME station-level coherence, fast-switch Bell roll-off and strong-curvature echo screens.

A.2 Allowed strength of the title question

The title asks “What Is Time?” The answer is intentionally an answer *within MMA-DMF*. It is not a claim that all alternatives have been eliminated. The allowed formulation is: time is modeled as causal relaxation of an active memory-bearing vacuum. The disallowed formulation would be: the final nature of time has been proved. The manuscript uses the former throughout.

A.3 Equation-spine dependencies

The equation spine has a dependency order. The retarded kernel cannot be interpreted until the GLE is stated. FDT closure cannot be interpreted until the retarded and even kernels are separated. Density-delay cannot be interpreted until the geometric relaxation law is stated. Bell roll-off cannot be interpreted until finite vacuum memory is defined. Voyager boundary timing cannot be interpreted until Shapiro, plasma and achromatic residual terms are separated.

B Extended Equation Notes

B.1 GLE as temporal law

The generalized Langevin equation is not merely a wave equation with a friction term. Its role in the temporal paper is to define the medium that updates. In ordinary geometric language, clocks trace worldlines and proper time is integrated along them. In the MMA-DMF language,

local systems relax through the active medium, and proper time is recovered as the screened projection of that process.

The memory integral is essential. If it were removed, the theory would still have a scalar field, but the strong temporal claim would weaken. Memory is what makes the vacuum history-sensitive. The FDT closure is also essential. If stochastic forcing were independent of drag, the model would become an adjustable noise model rather than a closed dissipative system.

B.2 Density-dependent mass and locality

Eq. (2) is the bridge from ontology to observable locality. Dense matter changes the effective mass and therefore the scalar response length. Because clocks, detectors and propagating signals are made of physical systems embedded in local environments, density becomes a timing variable. This is the model’s reason for rejecting a universal master clock while still retaining operational clock comparisons.

The Yukawa smoothing in Eq. (4) prevents the density law from being a pointwise slogan. Station-level timing cannot be predicted by an arbitrary single number for a location. It needs an effective density, integrated over a response kernel. This is why future GNOME analysis requires station metadata, shielding, overburden and calibration information rather than a public summary alone.

B.3 FDT closure and the arrow

The pair of kernels in Eq. (5) and Eq. (6) is the cleanest mathematical place where the paper avoids a common ambiguity. The retarded kernel is causal. The even kernel is a correlation object. The first one belongs in the memory integral; the second one belongs in equilibrium normalization. Confusing them would blur the arrow-of-time argument.

With the separation maintained, the model has a bounded statement: temporal directionality is built into the causal response law, while stochastic fluctuations remain thermodynamically closed. This is stronger than an analogy to entropy and weaker than a claim of final experimental proof.

B.4 Clock recovery as a constraint

A model that fails the Einstein limit fails immediately. The clock section is therefore written as a constraint rather than as a rhetorical comparison. GNSS and laboratory clock behavior are not weak confirmations; they are requirements that the screened limit must absorb. The Bear-Hug recovery section makes this explicit by stating the recovery equations before discussing boundary anomalies.

C Expanded Layer Notes

C.1 Bell sector

The Bell layer has two mathematically distinct parts. First, the CH/Eberhard route uses the inequality structure in Eq. (23). This layer is retained as `PASS_STRONG_EXTERNAL_BELL_LAYER` with an aggregate proxy near 13.4σ . Second, the all-trials CHSH bridge uses Eq. (25), yielding

$S_{\text{CHSH}} = 2.000029 \pm 0.000008$ with $z \approx 3.63$. The second number is deliberately bounded: it is not a NIST-published CHSH statistic and not a 5σ CHSH claim.

The next-phase result is Eq. (26). It is stronger as a future test than as a retrospective explanation. A controlled switching-frequency experiment can ask whether correlations degrade toward the classical bound when switching outruns the modeled contextual response. A null result across the predicted roll-off region would be a clean stress test.

C.2 High-energy sector

The high-energy sector is strongest when read as a convergence layer, not as a single-object claim. FRB negative drift supplies the population-level sign asymmetry. Magnetars connect that sign behavior to compact-source timing. NuSTAR and IXPE provide high-energy timing and polarization contexts. NASA high-energy products provide instrument and mission constraints. The retained status is therefore `PASS_HIGH_ENERGY_CONVERGENCE`, with the boundary that no exclusive scalar detection is claimed.

The FRB values are retained because they are numerically specific: 421 drift rows, 370 robust-filtered rows, median drift $-1130.804 \text{ MHz ms}^{-1}$, clean negative fraction 0.991892, sign-test probability 3.51×10^{-105} , slope 2.428, and $R^2 = 0.344$. The sign is the main temporal quantity. The scaling is a secondary bridge. The layer would weaken if a larger, cleaner population erased the sign asymmetry or reversed the slope under the same filtering logic.

C.3 GNOME public products and station-level gate

GNOME has a public layer and a station-level timing gate. The public layer is already useful: it includes event summaries, sensitivity surfaces, ALP/domain-wall bounds and station-noise products. The station-level gate remains different: it requires time streams, calibrations, station metadata and local density or shielding information.

This distinction is the scientific boundary. Public products can show detector landscape and sensitivity. They cannot test Eq. (12) and Eq. (14) at station level. The decisive question is whether density-delay correction improves coherent negative-chirp likelihood relative to flat simultaneity.

C.4 Voyager boundary stack

Voyager is similarly split. LECP identifies the heliopause transition context and gives a strong boundary layer. PDS/SPICE/SEDR gives ephemeris and trajectory context. DSN residual or operational support records are needed for observed-minus-computed timing. Without the last layer, there is no DSN residual proof. With the last layer, Eq. (32) becomes decisive because an achronatic residual can be separated from plasma dispersion.

The paper avoids three claims. It does not claim NASA concealment. It does not claim GR falsification without residuals. It does not treat missing DSN records as missing theory support. It treats them as the next required external adjudication layer for the strongest boundary-timing prediction.

C.5 Technion, HETDEX, GNSS and GW layers

The Technion/topological phase layer is a causality-safe bridge: apparent superluminal phase-feature motion does not imply superluminal signalling. The HETDEX/Ly α layer is a high-redshift baryonic-web bridge, not a direct local-time proof. The GNSS source layer is a screened-limit clock/orbit source layer, with a physical residual fit left as a next phase. The GW echo screen is a negative/screening result, not an echo detection.

D Supplementary Equation List

Table 10: Supplementary list of manuscript equations and their temporal roles.

Equation family	Formula link	Temporal role	Main test or layer
Relativistic GLE	(1)	Defines local causal update of the active vacuum.	Formal temporal ontology
Effective mass	(2)	Converts density into response length and clock sensitivity.	Clock and density tests
Effective density	(3)	Replaces point density by a physically averaged density field.	GNOME, laboratory density modulation
Yukawa smoothing	(4)	Normalizes the scalar density convolution.	Station-level timing
Retarded kernel	(5)	Enforces causal memory.	Arrow of time
Even kernel	(6)	Supplies FDT autocorrelation normalization without advanced response.	Kernel/FDT closure
FDT closure	(7)	Locks noise to dissipation.	Thermodynamic stability
Noise spectrum	(8)	Gives frequency and spatial ultraviolet regulation.	Stochastic bath closure
Geometric relaxation	(9)	Maps density to local response rate.	Local time
Relaxation derivative	(10)	Shows decreasing relaxation time with increasing density.	Density-delay sign checking
Density delay	(11), (12)	Defines station-to-station timing offsets; the second formula is the low-density approximation.	GNOME station gate
Negative chirp	(13)	Gives dissipative frequency drift sign.	FRB and transient tests
Network coherence	(14)	Defines likelihood structure for station-level events.	GNOME station gate
Clock limit	(15)	Recovers screened Einstein-limit clock ratios.	GNSS and clock tests
Proper time	(16)	Retains local Lorentz kinematics.	Einstein-limit recovery
Shapiro delay	(17)	Separates standard geometric delay.	DSN boundary timing
Plasma delay	(18)	Separates dispersive plasma terms.	DSN S/X adjudication
Einstein absorption	(19)	States dense screened recovery of GR observables.	Bear-Hug closure
Carrier phase	(20)	Links scalar state to local phase accumulation.	Quantum-time emergence
Wavefunction map	(21)	Interprets effective wavefunction as envelope/statistical map.	Quantum bridge
Context density	(22)	Encodes finite memory and density-conditioned contextual response.	Bell interpretation
CH/Eberhard	(23)	Defines the strong Bell-sector route.	Bell layer
CHSH bridge	(25)	Defines derived all-trials bridge.	Bell bridge
Dynamic roll-off	(26)	Predicts frequency-dependent contextual loss.	Controlled fast switching
FRB drift	(27)	Applies relaxation chirp to population drift.	High-energy timing
Robust filtering	(28)	Defines MAD filtering for drift rows.	FRB population layer
Sign test	(29)	Quantifies negative-drift sign asymmetry.	FRB population layer
ALP bound context	(30)	Maps public bound products into detector sensitivity context.	GNOME public layer

Equation family	Formula link	Temporal role	Main test or layer
Voyager boundary	(31)	Defines observed-minus-model boundary residual target.	Voyager DSN gate
DSN achromaticity	(32)	Distinguishes achromatic boundary residuals from plasma dispersion.	Voyager S/X gate
Phase singularity	(33)	Separates phase-feature motion from signal propagation.	Technion bridge
$\text{Ly}\alpha$ bridge	(34)	Connects baryonic web density to thermodynamic timing context.	HETDEX bridge
GW dispersion	(35)	Places strong-curvature timing in GW screen screened/dispersive form.	
GW echo	(36)	Gives mass-scaled echo target for future GW matched filtering.	next phase

E Expanded Test-Layer Descriptions

This appendix expands the compact inventory into purpose, setup, method, result, status and relation to the temporal ontology. It is included because the main paper should read as a manuscript, while the supporting test record must remain sufficiently complete for review.

E.1 Temporal closure audit

Purpose. The temporal closure audit asks whether the stated equation spine is sufficient to express a local theory of time without appealing to a universal clock. **Setup.** The required ingredients are the GLE, density-dependent mass, retarded memory, FDT closure, relaxation time and screened clock limit. **Method.** The audit checks whether each temporal claim can be linked to an equation and whether every equation has a falsification route. **Result.** The core temporal ontology is retained as a core formal claim. **Relation to time.** Time is not introduced as an independent object; it is the ordering generated by Eq. (1) under Eq. (5) and Eq. (7).

E.2 Arrow-of-time negative-chirp layer

Purpose. The negative-chirp layer tests the sign of dissipative relaxation. **Setup.** A relaxation event is modeled by Eq. (13). **Method.** The admissible template requires $d\nu/dt < 0$ for the relevant relaxation regime. **Result.** Negative chirp is retained as the operational sign of the temporal arrow inside the scalar-relaxation template. **Boundary.** This does not claim that all astrophysical chirps are scalar-vacuum events. It states the sign expected for the model’s relaxation channel.

E.3 No-master-clock delay layer

Purpose. This layer asks whether distributed sensors should be synchronized by a flat arrival-time assumption or by local relaxation delays. **Setup.** Station-dependent density enters through Eq. (9), Eq. (11) and Eq. (12). **Method.** A future station-level likelihood should compare density-delay correction to flat simultaneity using Eq. (14). **Result.** The layer remains a decisive external gate for GNOME station streams. **Relation to time.** A positive result would support the local-time thesis directly.

E.4 Bell CH/Eberhard closure layer

Purpose. The Bell layer asks whether the framework’s contextual sector has a strong external bridge rather than a loose analogy. **Setup.** The CH/Eberhard statistic is Eq. (23). **Method.** The retained route uses official-code-derived CH/Eberhard logic and keeps the all-trials CHSH bridge separate. **Result.** The CH/Eberhard route is retained as `PASS_STRONG_EXTERNAL_BELL_LAYER` with aggregate proxy near 13.4σ . **Boundary.** The CHSH bridge is not a NIST-published CHSH statistic.

E.5 Derived all-trials CHSH bridge

Purpose. This bridge connects all-trials outcome accounting to a CHSH-style number without overclaiming. **Setup.** Eq. (25) defines the bridge. **Method.** The derived all-trials calculation gives $S_{\text{CHSH}} = 2.000029 \pm 0.000008$ with $z \approx 3.63$. **Result.** The status is `PASS_COMPUTED_BOOTSTRAPPED`. **Boundary.** It is not a 5σ CHSH claim and not a substitute for the CH/Eberhard route.

E.6 Dynamic contextuality roll-off

Purpose. The dynamic roll-off layer turns contextuality into a temporal prediction. **Setup.** Eq. (26) defines the switching-frequency response. **Method.** A controlled Bell test would vary ν_{switch} and compare the measured trend to the finite-memory curve. **Result.** This is a next-phase collaboration test. **Falsification route.** Frequency-independent violation across the predicted roll-off region would weaken the finite-memory contextuality extension.

E.7 FRB robust negative-drift layer

Purpose. The FRB layer asks whether a high-energy population exhibits the sign asymmetry expected from negative-chirp relaxation. **Setup.** The retained population has 421 drift rows and 370 robust-filtered rows. **Method.** Eq. (28) filters outliers and Eq. (29) quantifies the clean negative fraction. **Result.** The clean negative fraction is 0.991892 with $p = 3.51 \times 10^{-105}$; the status is `PASS_HIGH_ENERGY_CONVERGENCE`. **Boundary.** It is not an exclusive scalar detection.

E.8 Magnetar and compact-source high-energy context

Purpose. This layer places FRB sign asymmetry inside compact-source timing rather than leaving it as a single radio-population observation. **Setup.** Magnetars, NASA high-energy products, NuSTAR and IXPE provide timing and polarization contexts. **Method.** The compact-source layer is interpreted through relaxation times, chirp signs and polarization or timing constraints. **Result.** It remains part of `PASS_HIGH_ENERGY_CONVERGENCE`. **Boundary.** Exact NuSTAR or IXPE numbers are not introduced where they are not present in the retained record.

E.9 GNOME public source layer

Purpose. The public GNOME layer defines detector landscape, sensitivity and noise context. **Setup.** The retained public products include 165 Fig. 2 rows, a 200×1001 sensitivity surface with 200,000 finite values, an ALP/domain-wall bound grid with 501,000 finite values, and a station-noise layer with 9 stations and 5753 intervals. **Method.** These products constrain

detector sensitivity and noise rather than station-level timing. **Result.** The status is `PASS_PUBLIC_SOURCE_LAYER`. **Boundary.** Station streams remain necessary for Eq. (14).

E.10 GNOME ALP/domain-wall bound layer

Purpose. This layer keeps the ALP figure without making an axion-exclusion claim. **Setup.** Public bound products are mapped into detector sensitivity context through Eq. (30). **Method.** The bound grids define detector reach and transient sensitivity. **Result.** The status is `PASS_GNOME_PUBLIC_ALP_BOUND_LAYER`. **Boundary.** Under MMA-DMF, an independent axion sector is not required as a fundamental input, but axions are not experimentally disproved by this manuscript.

E.11 Voyager LECP heliopause layer

Purpose. The LECP layer identifies boundary context at the heliopause. **Setup.** Retained values are 602,986 normalized rows, 257 computed channel steps, top robust $|z| = 365.375$, V1 boundary context near 2012-08-25 and V2 boundary context near 2018-11-05. **Method.** The layer tracks flux and channel-step changes near boundary epochs. **Result.** The status is `PASS_STRONG_BOUNDARY_CONTEXT`. **Boundary.** LECP is not DSN observed-minus-computed residual proof.

E.12 Voyager PDS/SPICE/SEDR ephemeris layer

Purpose. This layer establishes trajectory and ephemeris context for any later DSN residual analysis. **Setup.** It retains 269 source files, 8/8 parsed tables, fixed SEDR time handling, fill/sentinel filtering, SPICE-SEDR consistency and closest-approach geometry context. **Method.** The layer checks whether the ephemeris context is usable for boundary timing. **Result.** The status is `PASS_PDS_EPHEMERIS_CONTEXT`. **Boundary.** It is not a DSN residual proof.

E.13 Voyager DSN records gate

Purpose. This is the decisive deep-space boundary-timing gate. **Setup.** Required records include radiometric Doppler/range residuals, operational support records, calibration metadata and orbit-determination context. **Method.** Eq. (31) subtracts GR, plasma, station and orbit terms; Eq. (32) checks whether any residual is achromatic. **Result.** The status is `OPEN_RECORDS_PENDING` and `DECISIVE_FALSIFICATION_PENDING`. **Boundary.** No claim of concealment or GR falsification is made.

E.14 GNSS source-product layer

Purpose. GNSS tests the screened clock recovery path. **Setup.** Clock and orbit products supply the standard relativistic timing environment. **Method.** Eq. (15) and Eq. (16) must absorb ordinary relativistic behavior before any physical residual is interpreted. **Result.** The status is `PASS_SOURCE_LAYER`. **Boundary.** A physical residual fit remains a next phase.

E.15 HETDEX/Ly α bridge

Purpose. This bridge connects high-redshift baryonic structure to density and thermodynamic timing context. **Setup.** The bridge uses Eq. (34). **Method.** Ly α intensity and baryonic web products are read as cosmological density-response context. **Result.** The status is PASS_COSMOLOGICAL_BRIDGE. **Boundary.** It is not a local-clock proof.

E.16 Technion/topological phase bridge

Purpose. This layer keeps phase-singularity dynamics causality-safe. **Setup.** Eq. (33) separates phase-feature velocity from signal velocity. **Method.** The bridge treats apparent superluminal phase motion as a correlation property of phase topology. **Result.** The status is PASS_CAUSALITY_SAFE_TOPOLOGICAL_BRIDGE. **Boundary.** No superluminal signalling is claimed.

E.17 GW echo screen

Purpose. The GW layer tests strong-curvature timing without claiming an echo detection. **Setup.** Eq. (35) and Eq. (36) define the screen. **Method.** The retained proxy checks mass-scaled timing context and screening behavior. **Result.** The status is SCREENING_RESULT_CLOSED. **Boundary.** A full matched-filter analysis with trials accounting remains future work.

E.18 Atom-interferometer density modulation

Purpose. This is a clean laboratory route for the density-dependent response. **Setup.** A controlled nearby mass distribution would modulate local density while an interferometer measures phase or contrast. **Method.** The prediction links Eq. (2), Eq. (9), Eq. (11) and Eq. (12). **Result.** The status is NEXT_PHASE. **Falsification route.** A null result below claimed sensitivity or a wrong sign after controls would weaken the density-response sector.

E.19 Full GW matched-filter analysis

Purpose. This next-phase test asks whether strong-curvature timing produces mass-scaled echoes or only screening-consistent nulls. **Setup.** It requires event-level strain, locked templates, off-source backgrounds, detector consistency and trials accounting. **Method.** Eq. (36) supplies the mass-scaled target. **Result.** The status is NEXT_PHASE. **Boundary.** The current manuscript does not claim a detection.

F Master Test Inventory

Table 11: Master chronological test inventory retained for the temporal paper. Long records are condensed into scientific purpose, method, result, status and formula link.

Date or sequence	Test/layer	Purpose and setup	Result and status	Formula links
2026-04-15	Temporal ontology synthesis	Defines the time thesis: local causal relaxation rather than universal background time.	Retained as core formal	(1), (5), (7)

Date or sequence	se-Test/layer	Purpose and setup	Result and status	Formula links
2026-04-15	Arrow-of-time negative-chirp theory test	Tests whether retarded memory and FDT imply decreasing relaxation frequency.	Negative chirp retained	(13)
2026-04-15	No-master-clock delay law	Uses density-dependent relaxation to define station-to-station timing offsets.	Retained as model pre-diction; station streams pending.	(9), (11), (12)
2026-04-16	Equation inventory	Consolidates GLE, mass, kernel, FDT, relaxation, clock and spine contextual equations.	Retained as formula	Multiple in Table 4.
2026-04-16	Equation-to-test cross-walk	Maps equations to observables and falsification routes.	Retained in tables and appendices.	Multiple
2026-04-16	Claim-boundary table	Separates core claims, interpretations, bridge layers and future work.	Retained in Table 2.	Multiple
2026-04-16	Paper skeleton	Establishes the journal-style section order and abstract boundaries.	Rewritten in the present manuscript.	Multiple
2026-04-17	Dynamic contextuality roll-off	Uses finite vacuum memory to predict a rate-dependent approach from $2\sqrt{2}$ to 2.	Consistency limits retained; experimental roll-off pending.	(26)
2026-04-17	Bell official-code CH/Eberhard route	Computes CH/Eberhard quantities using official-code-derived logic.	PASS_STRONG_EXTERNAL (23) _BELL_LAYER ; aggregate proxy $\sim 13.4\sigma$.	
2026-04-17	Derived all-trials CHSH bridge	Uses frozen click/no-click all-trials mapping and bootstrap diagnostics.	$S_{\text{CHSH}} = 2.000029 \pm 0.000008$; $z \approx 3.63$; not a NIST-published CHSH statistic.	(25)
2026-04-18	FRB drift/DM summary	Aggregates FRB drift entries and dispersion context.	421 drift rows; robust filtering required.	(27), (28)
2026-04-18	FRB robust negative-drift layer	Applies MAD filtering and sign test to clean drift rows.	370 clean rows; negative fraction 0.991892; $p = 3.51 \times 10^{-105}$; PASS_HI_GH_ENERGY_CONVERGENCE .	(13), (29)
2026-04-18	Magnetar/high-energy archive extension	Places FRB sign asymmetry in compact-source timing and polarization context.	Retained as high-energy convergence; no exclusive scalar detection.	(27), (35)
2026-04-19	GNOME public positive-event layer	Summarizes public Fig. 2 products.	165 rows; retained as public source layer.	(14)
2026-04-19	GNOME sensitivity-surface layer	Summarizes Fig. 3 sensitivity grid.	200,000 finite grid values; PASS_PUBLIC_SOURCE_LAYER .	(14)
2026-04-19	GNOME ALP/domain-wall bound layer	Summarizes Fig. 4 bound products.	501,000 finite values; axion sector not required as fundamental input under MMA-DMF.	(30)
2026-04-19	GNOME station-noise layer	Summarizes public Data noise products.	Extended 9 stations; 5753 intervals; ~ 179.91 station-days; median ~ 3.32 pT.	(14)
2026-04-19	GNOME station-level timing gate	Requires station streams, sampling rates and local density metadata.	DECISIVE_FALSIFICATION_PENDING .	(12), (14)
2026-04-20	GNSS clock/orbit source layer	Parses clock/orbit products and establishes a residual-fit pathway.	PASS_SOURCE_LAYER ; physical residual fit next phase.	(15), (16)
2026-04-20	Bear-Hug limit recovery	Einstein-Clocks Checks that dense screened clocks and geometry recover GR timing.	CLOSED ; required baseline.	(15), (17)

Date or sequence	se-Test/layer	Purpose and setup	Result and status	Formula links
2026-04-20	Voyager LECP heliopause transition	Uses LECP boundary windows around V1/V2 heliopause text.	602,986 rows; 257 computed channel steps; top $ z = 365.375$; PASS_STRONG_BOUNDARY_CON TEXT.	(31)
2026-04-20	Voyager PDS/SPICE/SEDR ephemeris	Parses PDS trajectory context and checks SEDR/SPICE consistency.	269 files; 8/8 tables parsed; PASS_PDS_EPHEMERIS_CONTEXT .	(31), (32)
2026-04-20	Voyager DSN residual gate	Requires radiometric or operational support records with observed-minus-computed residuals.	OPEN_RECORDS_PENDING (32), (18)	
2026-04-21	HETDEX/Ly α bridge	Connects high-redshift baryonic web and Ly α observables to density/thermodynamic timing.	PASS_COSMOLOGICAL_BRIDGE (34)	
2026-04-21	Technion/topological phase bridge	Interprets apparent superluminal phase-singularity motion as causality-safe phase dynamics.	PASS_CAUSALITY_SAFE_TOPOLOGICAL_BRIDGE .	(33)
2026-04-21	GW150914 echo-screen proxy	Uses mass-scaled echo window as strong-curvature screen.	SCREENING_RESULT_CLOSER ; no detection claimed.	(36)
2026-05-02	Kernel/FDT consistency	Checks retarded/even separation and FDT factor.	Consistency closed.	(5), (6), (7)
2026-05-02	Density-delay derivative check	Verifies $df_{\text{delay}}/d\rho < 0$.	Consistency closed.	(12)
2026-05-02	CHSH static convention	Confirms static contextual convention near $2\sqrt{2}$ with fixed κ_* .	Consistency closed; does not replace Bell data.	(22), (26)
2026-05-02	Dynamic CHSH roll-off limits	Checks low-rate and high-rate limits.	Consistency closed.	(26)
Next phase	Atom-interferometer density modulation	Controlled mass-density modulation near a long-baseline interferometer.	NEXT_PHASE .	(2), (9)
Next phase	Controlled fast-switch Bell experiment	Variable switching frequency with locked metadata and outcome streams.	PENDING_COLLABORATION_REVIEW .	(26)
Next phase	Full GW matched-filter echo analysis	Event-level templates, source backgrounds and trials accounting.	NEXT_PHASE .	(36)

G Additional Claim-to-Falsification Map

Table 12: Supplementary claim-to-falsification map.

Claim	Main equation links	Strongest supportive layer	Direct route	weakening
Time as local causal relaxation	(1), (5), (7)	Formal equation spine and kernel/FDT consistency	Instability, response, or inability to close noise with dissipation	acausal

Claim	Main equation links	Strongest supportive layer	Direct route	weakening
Density-dependent local time	(9), (11), (12)	GNOME station-level prediction and density-modulation laboratory route	Station-level flat density modulation gives wrong sign	streams flat simultaneity or laboratory density modulation gives wrong sign
Negative-chirp relaxation	(13), (27)	FRB sign asymmetry and high-energy timing convergence	Clean erase asymmetry or validated positive survive controls	populations negative sign or scalar-like chirps survive controls
Einstein-limit clocks	(15), (16), (17)	GNSS source layer and Bear-Hug recovery	Failure to recover screened clock behavior or residuals requiring incompatible parameters	
Bell contextual memory	(23), (25), (26)	CH/Eberhard closure and derived CHSH bridge	Reanalysis failure or no dynamic roll-off where predicted	
Voyager boundary timing	(31), (32), (18)	LECP/PDS boundary and ephemeris context	DSN records show no achromatic boundary component after corrections	
GNOME detector landscape	(14), (30)	Public source, sensitivity, bound and noise products	Station-level timing rejects density-delay negative-chirp coherence	
Strong-curvature screen	(35), (36)	GW echo screen as negative/screening result	Full matched-filter search contradicts fixed template expectations	

H Supplementary Notes on Remaining Gates

H.1 NASA/Voyager DSN gate

A useful DSN gate would include S/X-band radiometric Doppler and range residuals, observed-minus-computed products, calibration records, station metadata, orbit-determination notes and refit history. The test is not whether the Voyager heliopause exists; that boundary is already part of the physical context. The test is whether an achromatic residual term survives standard corrections at the boundary.

A supporting result would show a boundary-correlated achromatic component that survives

plasma, station, orbit, refit and time-shift nulls. A weakening result would show that the apparent timing structure is fully absorbed by plasma dispersion, station effects, trajectory refit or ordinary clock/orbit corrections.

H.2 GNOME station-level timing gate

A useful GNOME timing gate would include Science Run 2 station streams, sampling rates, clock metadata, calibration, local noise, station shielding, environmental monitors and density or overburden proxies. The test is not whether GNOME has sensitivity to exotic transients in general. The test is whether Eq. (14) improves when density-delay and negative-chirp constraints are applied.

A supporting result would show coherent negative-chirp events with improved likelihood under density-delay correction. A weakening result would show equal or better coherence under flat simultaneity, robust positive chirps passing controls, or absence of events in a claimed sensitivity region.

H.3 Next-phase laboratory gates

The controlled fast-switch Bell roll-off, atom-interferometer density modulation, GNSS physical residual fit and full GW matched-filter echo analysis are not required to define the temporal ontology. They are useful next-phase discriminants. Their role is to increase or decrease confidence in specific sectors without confusing formulation with validation.

I Supplementary Numerical Summary

The numerical values retained in the manuscript are deliberately status-labeled. FRB values belong to `PASS_HIGH_ENERGY_CONVERGENCE`. GNOME public products belong to `PASS_PUBLIC_SOURCE_LAYER` or `PASS_GNOME_PUBLIC_ALP_BOUND_LAYER`. Voyager LECP and PDS values belong to `PASS_STRONG_BOUNDARY_CONTEXT` and `PASS_PDS_EPHEMERIS_CONTEXT`. Bell CH/Eberhard belongs to `PASS_STRONG_EXTERNAL_BELL_LAYER`. The CHSH bridge belongs to `PASS_COMPUTED_BOOTSTRAPPED`. GW echo screening belongs to `SCREENING_RESULT_CLOSED`.

The status labels are part of the result. They prevent the manuscript from making a single universal validation claim out of heterogeneous evidence.

J Figure Placement Rationale

All figures are retained and placed where their scientific function is strongest:

1. the density-law figure appears in the mathematical core because it supports the local-time mechanism;
2. the Bear-Hug figure appears in the Einstein-limit section because it separates closed recovery from open boundary gates;
3. the Bell and roll-off figures appear in the Bell section because they separate the completed CH/Eberhard layer from the dynamic future test;

4. the FRB figures appear in the high-energy section because FRB drift is the population-level negative-chirp layer;
5. the GNOME and ALP figures appear in the GNOME section because they distinguish public detector products from the station-level timing gate;
6. the Voyager figures appear in the Voyager boundary stack because they are LECP context, not DSN residual proof;
7. the GW figure appears in the strong-curvature section because it is a screening result, not an echo detection.

K Acceptance Criteria and Failure Modes

This appendix states the acceptance logic in the form a reviewer can use to separate a completed layer from a future gate. Each item has three parts: what would count as support, what would count as weakening or falsification, and what cannot be claimed from the layer alone.

K.1 Core temporal formulation

Support. The formulation is supported when the GLE, kernels, FDT closure, density law and clock limit are mutually consistent and linked to explicit observables. The model is strongest when Eq. (1), Eq. (5), Eq. (6), Eq. (7), Eq. (9) and Eq. (15) are read as one spine rather than separate motifs. **Weakening.** The core would weaken if the same observable required incompatible kernels, incompatible density scalings or an uncoupled noise sector. **Boundary.** A mathematically consistent temporal spine is formulation, not external proof.

K.2 Arrow of time

Support. The arrow is supported internally by causal retardation and FDT closure. Externally, it is supported by relaxation classes that follow negative-chirp behavior under the same template logic. **Weakening.** A robust scalar-like positive-chirp population passing the same controls would stress the arrow sector. **Boundary.** The paper does not claim that the thermodynamic arrow has been experimentally reduced to one signal class. It states the sign expected for the modeled relaxation channel.

K.3 Density-delay timing

Support. A station-level network event supports the density-delay layer only if density-corrected timing improves the likelihood relative to flat simultaneity and if local noise, calibration and station metadata are controlled. **Weakening.** Equal or better likelihood under flat simultaneity weakens the local-time prediction. **Boundary.** Public summaries and sensitivity products identify where the test is possible; they do not by themselves close the timing gate.

K.4 Einstein-limit recovery

Support. Einstein-limit recovery is supported when dense screened environments absorb scalar corrections into the standard clock and geometric observables. In practice this means that GNSS, Shapiro delay, ordinary proper time, gravitational redshift and local Lorentz kinematics are

recovered before any residual interpretation is attempted. **Weakening.** Failure to reproduce known screened timing behavior would be fatal to the clock section. **Boundary.** Recovering the Einstein limit is necessary but not sufficient to prove the deeper ontology.

K.5 Bell-sector interpretation

Support. The Bell sector has a completed CH/Eberhard layer and a bounded computed CHSH bridge. It becomes a temporal test only when switching frequency is varied and compared to Eq. (26). **Weakening.** A controlled fast-switching experiment showing no roll-off in the predicted region would weaken dynamic contextuality. **Boundary.** The derived bridge cannot be written as a published 5σ CHSH result.

K.6 FRB/high-energy interpretation

Support. FRB negative drift supports high-energy timing convergence because the sign asymmetry is large and the filtered sample remains strongly negative. Compact-source contexts extend the same relaxation ontology beyond one radio-population layer. **Weakening.** A larger clean sample that removes the negative sign asymmetry under the same robust filtering would weaken the layer. **Boundary.** The layer is not an exclusive scalar detection and does not eliminate conventional astrophysical modeling.

K.7 GNOME public products

Support. GNOME public products support the detector landscape, source-data, sensitivity, bound and station-noise context. They also define realistic thresholds for future station-level analysis. **Weakening.** If future station streams show no density-delay improvement or no negative-chirp coherence in the claimed sensitivity region, the timing prediction weakens. **Boundary.** Public products are not station-level timing validation.

K.8 GNOME ALP/domain-wall context

Support. The ALP/domain-wall bound figure is useful because it maps the detector landscape. **Weakening.** It would weaken only the sensitivity-context interpretation if the public bound products were misread or incompatible with detector response. **Boundary.** MMA-DMF does not require axions as fundamental inputs, but this paper does not experimentally disprove axions.

K.9 Voyager LECP and PDS context

Support. LECP supports heliopause boundary context; PDS/SPICE/SEDR supports ephemeris context. Together they establish where a DSN residual test would be meaningful. **Weakening.** These layers would weaken if the boundary epochs or ephemeris context were grossly incompatible with the retained transition interpretation. **Boundary.** Neither layer is observed-minus-computed DSN proof.

K.10 Voyager DSN residual gate

Support. A supportive DSN result would show a heliopause-correlated achromatic residual component after plasma, station, orbit and refit controls. **Weakening.** A complete DSN record showing no achromatic component with adequate sensitivity would weaken the boundary-timing prediction. **Boundary.** Absence of public residual records is not evidence for the theory and not evidence against NASA. It is only a statement of missing external adjudication material.

K.11 GNSS residual fit

Support. A supportive GNSS next phase would show residual patterns compatible with screened local scalar response after standard corrections. **Weakening.** Inability to recover standard timing or residuals requiring incompatible parameters would weaken the sector. **Boundary.** GNSS source products alone are a source layer, not a completed residual fit.

K.12 HETDEX/Ly α bridge

Support. The bridge supports the idea that high-redshift density and baryonic web structure can be connected to thermodynamic timing context. **Weakening.** It would weaken if the bridge required a transfer function incompatible with the rest of the equation spine. **Boundary.** It is not a local clock experiment.

K.13 Technion/topological phase dynamics

Support. The bridge is useful because it illustrates how apparent superluminal phase features can remain causality-safe. **Weakening.** It would weaken only if the phase-feature interpretation were treated as signal propagation. **Boundary.** The paper does not claim superluminal communication.

K.14 GW echo screen

Support. The GW screen is supportive if fixed templates produce either mass-scaled echoes or screening-consistent nulls under proper trials accounting. **Weakening.** Persistent absence in a region where the model predicts detectability would weaken the strong-curvature timing branch. **Boundary.** The present manuscript does not claim an echo detection.

L Reviewer-Facing Summary of the Manuscript Hierarchy

The manuscript uses four structural safeguards. First, the abstract is ontological and equation-centered. Second, the main text foregrounds the answer to the title question before discussing convergence layers. Third, Bell, FRB, GNOME and Voyager are given different evidential roles instead of being flattened into one support list. Fourth, the missing NASA/Voyager and GNOME records are treated as external falsification gates rather than as missing formulation.

The resulting hierarchy is direct. The thesis is stated first: time is causal relaxation of an active memory-bearing vacuum. The equation spine follows. Only after that does the paper introduce the convergence map. This ordering keeps the scientific boundaries explicit while preserving the central ontological argument.

M Compact Roadmap for Independent Review

An independent reviewer can assess the manuscript in the following order:

1. Check whether Eq. (1) through Eq. (7) form a consistent memory-and-FDT dynamics.
2. Check whether Eq. (9), Eq. (11) and Eq. (12) really support local timing rather than a master clock.
3. Check whether Eq. (15), Eq. (16), Eq. (17) and Eq. (19) protect the Einstein limit.
4. Check whether the Bell section separates CH/Eberhard from the derived CHSH bridge.
5. Check whether the high-energy section keeps FRB values and boundaries together.
6. Check whether GNOME public products are separated from station-level timing streams.
7. Check whether Voyager LECP/PDS context is separated from DSN residual proof.
8. Check whether all tables link back to manuscript equations rather than to external non-manuscript labels.
9. Check whether the final conclusion states formulation now and external falsification next.

This roadmap is intentionally falsification-oriented. The manuscript is not asking the reader to accept the ontology by rhetorical breadth. It asks the reader to follow the equation spine and then judge the gates.

N Extended Closing Note

The strongest form of the paper is neither a short philosophical essay nor a flat catalogue. It is a layered scientific manuscript. The ontology is one layer. The equations are a second. The completed convergence and bridge layers are a third. The two decisive external record gates are a fourth. Keeping these layers separate is what allows the title question to remain strong without making the text overclaim.

O Final Manuscript Checklist

The final manuscript uses only the name MMA-DMF for the framework. Equations are cited by manuscript numbers. Tables link to equations through equation references rather than through non-manuscript labels. Figures are placed in the scientific sections where they are interpreted. References appear after the supplementary material. Pages are kept in portrait orientation. The text separates completed convergence layers from the two decisive external record gates. The conclusion states that the framework is formulated now and that NASA/Voyager and GNOME records are required for independent external falsification of the strongest boundary-timing predictions.

The checklist matters because the paper's title is ontological. A paper titled "What Is Time?" must not read like a status dashboard. It must give the answer, then give the equations, then give the tests. The expanded version follows that order: time is causal relaxation; the equation spine defines it; the Einstein limit is recovered; Bell and high-energy sectors are completed

bridge or convergence layers; GNOME and Voyager identify the decisive timing gates; and the appendices preserve the detailed inventory without diluting the thesis.

P Supplementary Conclusion

The expanded record supports a focused manuscript rather than a shortened one. The paper is strongest when it says exactly what is closed, exactly what is a bridge, and exactly what remains externally decisive. The time ontology is formulated now. The missing NASA/Voyager and GNOME records decide the strongest boundary-timing predictions later.

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